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**Fitness and novelty in evolutionary
reinforcement learning**

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Title: Fitness and novelty in evolutionary reinforcement learning

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Abstract: Novelty is a novel approach to modelling selection criteria in evolutionary algorithms and has been proven as viable technique of avoiding pitfalls of false optima in tasks abundant with them, such as solving mazes. Rather than closing the topic however, this finding opened other problems to explore: How to properly apply novelty in tasks that yield slightly better to conventional approaches? How to properly model behavioural space necessary for novelty computation? In this thesis we investigate use of novelty in selected reinforcement learning tasks, it's combinations with classical fitness and propose behaviour space models for the respective RL tasks.

Keywords: evolution novelty fitness behavioural space

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Introduction

Evolutionary algorithms are a class of nature based optimisation algorithms predicated on the idea of "survival of the fittest" using the optimisation objective as "fitness", guiding the algorithm's search. However for complex objectives direct use of fitness might clog the algorithm in a false, local, optima, unable to reach true global optima. To solve this problem a variety of approaches have been suggested such as archiving, where variety of individuals is kept by an archive. Or novelty search where fitness is replaced by "novelty", a difference in behaviour from rest of the population. Particularly novelty has been shown to work in constrained reinforcement learning tasks such as mazes. The notion of novelty for reinforcement learning is being even generalised outside of the evolutionary algorithms [citation here]. On the other hand there is a number of tasks where combination with fitness might enhance the novelty search, suggesting local competition like MAP-ELites or Dominated Search or other k-neighbours algorithm. These techniques however tend to be expensive for computation.

In this thesis we will characterise two reinforcement learning environments, Carpole and Lunar lander, suggest interpretable behavioral representation for them and apply different evolutionary algorithm techniques, such as differential evolution, itself being a mechanism to introduce controlled diversity into the population. We will compare several lightweight applications of novelty combined with fitness, showing their viability and pitfalls on the tasks at hand and suggest best approach for each of the environments. In the chapter 1 we will introduce terminology pertaining to agents we will be optimising. In chapter 2 we will introduce the aforementioned reinforcement learning tasks and detail the optimising algorithms in chapter 3. In chapter 4 we will review background of the experiments and finally in chapter 5 we will discuss the results of our experiments.

1. Introduction to Artificial Agents

To understand the reasoning guiding this work we have to first look at the fundamental subject of our efforts - the artificial agents.

We will look at their structure, guiding principles, practical implementation and nor last nor least, their use.

1.1 Agents and environment

All agents exist in environments that provide context to their agency. An agent gets information about the **environment** it find itself in (i.e. **percepts**) through **sensors** and acts on the environment through **actuators**. Both actuators and sensors are usually characteristics of the agent. Another characteristic of an agent would be setup of its internal machinations.

Definition 1 (Russell and Norvig [2021]). *Let E be environment where agent G is located. G is equipped with actuators A and sensors P and so we can identify $obss_P(E)$, a space of all possible sequences of percepts available to G within E and $A(E)$ space of available actions in E . Then function $G : obss_P(E) \rightarrow A(E)$ is called an agent function, defining behaviour of the agent G in environment E .*

We illustrate relationship of these concepts in the diagram 1.1. Note that provided definition of agent policy is characteristic to deterministic agents. For agents in stochastic environments we need to reframe the agent function as probabilistic policy. We will discuss this reframing further in section 2.1.

Because we want our agents to solve some tasks in their respective environments with certain efficiency meaning we want them to be intelligent, we need to define a notion of rationality that can help us better capture what we mean by intelligence. First however we need to measure how useful or detrimental a situation can be for our agent.

Definition 2 ([Russell and Norvig, 2021]). *Be that E is an environment and $seq(E)$ is a space of all possible sequences of states within E . Then function $f : seq(E) \rightarrow \mathbf{R}$ is a performance measure of some agent operating in E .*

Definition 3 (Russell and Norvig [2021]). *For each possible percept sequence, a rational agent should select an action that is expected to maximise its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.*

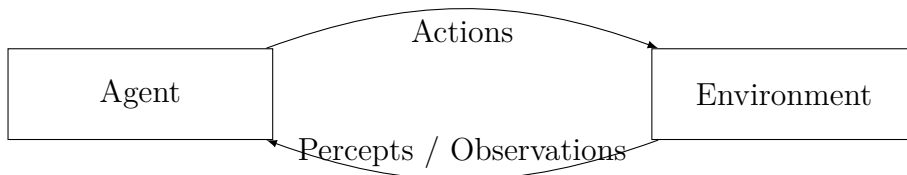


Figure 1.1: Agent and environment relationship

We should note that in the real-world applications the rationality of an agents can often be limited by their capacity to store the evidence gathered from percepts, by the lack of their built-in knowledge or the computational capacity to estimate action sequence that can be expected to achieve global maximum of the agent's performance measure.

This is often characterised as a capacity to form *belief*. With that we expand on the notion of agent policy In this thesis we are more concerned with the properties of the evolutionary algorithms rather than evolution of complex agents. Therefore we consider only *reflexive agents*.

Definition 4 ([Russell and Norvig, 2021]). *An agent is called simple reflex agent if it takes into account only the current percepts, never keeping any history or state information during it's run.*

1.2 Neural Network Agent

The concept of neural networks (NN) comes from an area of machine learning called deep learning that was conceived to attempt to model biological neuron activity. Since it has become one of the most successful branches of machine learning (Russell and Norvig [2021]). The neural networks consist of neurons organised in layers.

Definition 5. *Let $W \in R^{m \times n}$ and $b \in R^n$ and let $f : R^n \rightarrow R$. Then we call W the weights of a neuron layer with n neurons, b the bias of the layer and f the activation function of the layer when given an input vector $x \in R^m$ we calculate the output of the layer as $f(W^T x + b)$. Two layers have directional connection when the output of first layer is used as input of the subsequent layer.*

The purpose of the activation function is to introduce exploitable non-linearity into the neural system. We recognise several basic activation function.

- sigmoid

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

- tanh

$$\tanh(x) = \frac{e^{2x} - 1}{e^{2x} + 1}$$

- ReLU

$$ReLU(x) = \max(0, x)$$

Definition 6 (Russell and Norvig [2021]). *A neural network where the connections between layers of neural network form an acyclic graph is called a feed-forward neural network.*

Because the feed-forward NN doesn't hold any state or holds onto any previously acquired results and only produces output for a given input, they are ideal for implementation of simple reflex agents. In this thesis we use feed-forward networks exclusively, so any further mention of neural network should be assumed to be feed-forward.

2. Environments

In this chapter we will discuss the environments we have chosen to use in our examination of evolutionary algorithms and the theoretical background that will help us analyse them.

2.1 Reinforcement learning

Reinforcement learning might be best introduced by description of the problem it aims to solve.

Definition 7 (Markov's decision problem). *Let P be a probability measure over the states of the environment. We define Markov's decision problem (MDP) by a four-tuple (S, A, T, r) where:*

- S is a set of all possible states of the system, or state space
- A is a set of all possible actions available in the system or action space
- T is a transition function:

$$\forall s', s \in S, a \in A : T(s, a, s') = P(s'|s, a)$$

- $r : S \times A \times S \rightarrow \mathbf{R}$ is reward function, $r(s, a, s')$; being the reward received when the execution of action a in state s leads to state s' .

To tie this into the information from chapter 1 $A = A(E)$, R is performance measure of an agent solving the MDP and the percepts have bijection to real states of the environment. This would not be the case for partially observable MDP.

Definition 8 (Kaelbling et al. [1998]). *Let P be a probability measure over the states of the environment. We define Partially Observable Markov's decision problem (POMDP) by a six-tuple (S, A, Ω, T, O, r) where:*

- S is a set of all possible (true) states of the system, or state space
- A is a set of all possible actions available in the system or action space
- Ω is a set of all possible observations on the system, or observation space
- $T : S \times A \times S \rightarrow [0, 1]$: is a transition function over true states.

$$\forall s', s \in S, \forall a \in A : T(s, a, s') = P(s'|s, a)$$

.

- $O : S \times A \times \Omega \rightarrow [0, 1]$.

$$\forall s' \in S, \forall a \in A, \forall o \in \Omega : O(s', a, o) = P(o|a, s')$$

or the probability of making observation o given that the agent took action a and landed in state s' .

- $r : S \times A \times S \rightarrow \mathbf{R}$ is reward function same as in MDP.

Now that we have characterised the environments we can apply to them concepts we have revied in the chapter 1. We begin by specifying the notion of percept sequence space from definition 1.

Definition 9 (Ross et al. [2014]). *Let Ω the space of all observations ($\Omega \cong S$ for MDP), A be the action space, $o_t \in \Omega$ be observation at time t and $a_t \in A$ action taken by the agent in time t . Then*

$$h_t = (a_0, o_1, \dots, a_{t-1}, o_t)$$

is a history at time t .

Definition 10 (Ross et al. [2014]). *Let P be probability measure over states of the environment S and $s_t \in S$ be the state of the environment at time t . Then function $b_t : S \rightarrow [0, 1]$ such that*

$$b_t(s) = P(s_t = s | h_t, b_0)$$

is called a belief state of an agent, where b_0 is the apriori belief state and h_t is history at time t .

We also need to reframe the agent function discussed in definition 1.

Definition 11 (Ross et al. [2014]). *Let P be a probability measure and A be an action space. Then function π defined as $\pi(b_t, a) = P(a | b_t)$ where $P(a | b_t)$ denotes probability of doing action a given belief state b_t is called agent policy.*

Definition 12 (Ross et al. [2014]). *Let r be a reward function, A an action space, T a transition function, $\gamma \in [0, 1)$ a discount factor and Π be a set of all possible agent policies, then*

$$\pi^* = \arg \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t \sum_{s \in S} b_t(s) \sum_{a \in A} \sum_{s' \in S} r(s, a, s') T(s, a, s') \pi(b_t, a) \mid b_0 \right]$$

is called optimal policy.

The goal of reinforcement learning is then to proximate optimal policy by trial and error, based on the received reward, as accurately as possible. This can be done utilising various online or offline methods such as Q-learning. In the literature and specifically in the cited Ross et al. [2014] the the definition 12 features γ as an discounting factor for the reward. This is a crucial parameter for reinforcement learning algorithms that require continuous rewards. However as we will discuss further in this chapter and chapter 3, in evolutionary algorithms we evaluate the policy as a whole and are only presented with final value calculated. Therefore it is not necessary to know or discover particular value of γ for the environments presented in this thesis.

Note that since in this thesis we are only concerned with simple reflexive agents, the belief state is not explicitly calculated statistic but is nonetheless reflected in the learned reactions to observation.

2.2 Gymnasium

We intend to set our agents into environments provided by Gymnasium project which emerged from it's predecessor OpenAI Gym project. Both of them Python libraries, trying to provide researches with benchmark reinforcement learning problems in form of simple games, sometimes even recreating old console games such as Air Raid or Backgammon.

While Gymnasium provides wide range of environments, quite a number of them if not most do not provide their outputs in structured form. Instead the resulting screen has to be processed by user defined visual processor to extract information crucial for completing the task.

This introduces a layer of complexity that is outside of the scope of this thesis. Therefore we have decided to narrow our selection only to environments which do provide structured output. We describe the loop of receiving and providing feedback between agent and environment in algorithm 1.

Algorithm 1: Environment Interaction Loop

```
Input: Environment  $\mathcal{E}$ 
Input: Number of episodes  $n$ 
Input: Observation aggregation functions  $agg$  and  $transform$ 
 $R = 0$ ;
for  $i = 0$  to  $n$  do
     $o \leftarrow \text{reset}(\mathcal{E})$ ;
     $O \leftarrow \text{reset}(O)$ ;
    while not terminated do
         $a \leftarrow \text{Agent}(o)$ ;
         $(o, r, \text{terminated}) \leftarrow \text{Environment}(a)$ ;
         $R \leftarrow R + r$ ;
         $O \leftarrow agg(O, o)$ ;
    end
end
return  $R/n, transform(O)$ ;
```

As mentioned in the introduction the selected environments are called Cartpole and Lunar Lander.

We will discuss the full purpose of *agg* and *transform* in the section 3.2.2 and section 4.3.2.

2.2.1 Cartpole

Cartpole is one of the most used benchmark environments in the Gymnasium library. It's a simple environment where the agent controls a cart rolling on flat surface with a pole on top. The goal is to keep the pole balanced on top of the cart. The reward is calculated every step depending on the angle of the stick. This means we have bounded total reward between 0 and 500 points.

Definition 13. *An agent that achieves score 500 in Cartpole is called solution or solving individual.*

Observation space

Cartpole observation space is expressed as a 4-dimensional vector. We delineate possible values of the vector in 2.1. Note that for Cartpole environment this observation space reflects full state of the environment in the given moment and therefore it could be characterised as MDP rather than POMDP.

Index	State Variable	Min	Max
0	Cart position	-4.8	4.8
1	Cart velocity	$-\infty$	∞
2	Pole angle	-0.418 rad	0.418 rad
3	Pole angular velocity	$-\infty$	∞

Table 2.1: Observation space of Cartpole

Action space

The action space is discrete with only two values possible as presented in the table 2.2. Note that we cannot chose to do nothing.

Action	Description
0	Push cart to the left
1	Push cart to the right

Table 2.2: Cartpole Action Space.

2.2.2 Lunar Lander

The other environment we have selected is Lunar Lander where the agent controls descend of a landing shuttle on rugged planetary (or the Moon's) surface with the help of oddly placed rockets. The points are subtracted for staying in the air or for getting the shuttle into a spin. Bonus points awarded for landing in the centre of the The shuttle has infinite fuel so the game ends only when the shuttle lands safely or crushes into the ground. Luckily we can at least limit the number of steps for which the environment is allowed to go on. For a neural network training this however poses a specific kind of problem. Firstly very narrow behavioural window that can achieve positive total reward. But mainly as opposed to the Cartpole environment (2.2.1), now the unsuccessful runs may take significantly longer time than the successful ones. This means experiments are particularly costly for slowly converging algorithms. The upper limit of sum total reward is 200, the lower would possibly be $-\infty$, depending on the length limitation imposed. We therefore differentiate between several stages of obtaining successful agent.

Definition 14. *An agent that achieves score 200 in Lunar lander is called solution or solving individual.*

Definition 15. *An agent that achieves positive score in Lunar lander is called well-behaved.*

Definition 16. *An agent that achieves average score over 100 in Lunar lander is called very well-behaved.*

Definition 15 stems from the fact that it can only be achieved if the agent does not crash the shuttle as doing so yields a penalisation of -100 points while not obtaining a boost of +100 points for safe landing, driving even a possible solution with score 200 to 0. Similarly for definition 16, it is highly unlikely to average score over 100 without at least once landing with perfect score 200.

Observation space

The observation space of Lunar Lander is represented as 8 dimensional vector. Delineation of its values can be found in the table 2.3. Note that while the observation space is quite descriptive it does not capture all the information about environment’s state directly such as wind velocity or proximity to the ground. Because the landscape (except for a landing pad in the centre) might be different in each episode, we can only infer their parameters from the observations (such as velocity and leg contact with ground) indirectly. For simplicity we do not allow wind in our episodes as is the default setting of the environment. This means that the distinction between observation and true state becomes most tangible in low altitudes where agent risks either collision with unexpected hill or infinite hovering above unperceived valley, forgoing the +100 point reward.

Index	State Variable	Min	Max
0	x position	-2.5	2.5
1	y position	-2.5	2.5
2	v_x velocity in x-axis	-10	10
3	v_y velocity in y-axis	-10	10
4	angle θ	-6.2831855	6.2831855
5	angular velocity	-10	10
6	left leg contact (boolean)	0	1
7	right leg contact (boolean)	0	1

Table 2.3: Observation space of Lunar Lander

Action space

Action space for Lunar Lander can come in two shapes. First is discrete where we the agent chooses a number representing one of the engines to ignite. Second is continuous where agent has continuous control over the ignition of the main engine and one of the side engines. This variant allows for more precision during the flight but at the same time makes the policy the agent needs to learn significantly more complex, requiring more resources during the learning phase. To preserve compatibility with Cartpole setup and considering feasibility of multiple experiments ahead we will work with the discrete version of the action space outlined in table 2.4.

Action	Description
0	Do nothing
1	Fire left orientation engine
2	Fire main engine
3	Fire right orientation engine

Table 2.4: Action space definition for the control system.

3. Evolutionary algorithms

The evolutionary algorithms (EA) are a class of optimisation methods modelled after our ideas of biological process of evolution. This chapter is based primarily on Introduction to Evolutionary Computing from Eiben and Smith [2003].

Definition 17. *General maximising optimisation problem takes the form:*

$$\begin{aligned} \text{maximise: } & f(x) \\ \text{subject to: } & g_i(x) \leq 0, \quad i = 1, \dots, m, \\ & h_j(x) = 0, \quad j = 1, \dots, p, \end{aligned}$$

where x is the solution vector, f is objective function, and g_i, h_j are constraints imposed on x

Although theoretically it could be also minimising, in this thesis we will not encounter the minimising optimisation and so our concern is mainly with the maximising case.

Because of this biological connection solution vectors are in the literature often referred to as individuals and we will refer to them this way henceforth as well. Alternatively when emphasising either the numerical form of the vector or semantic interpretation of this form beyond the optimisation algorithm the individual may be refer to as either a **genotype** or **phenotype** respectively.

Group of individuals representing candidate solutions at a given time is referred to as a population. In general case evolutionary algorithms then evolve the population by applying three operations (operators) to it in a loop until either the population converges to optimum or the limit on number of steps is reached.

Definition 18. *Let f be objective function in minimisation (resp. maximisation) problem. We call an individual i more successful than individual j if value $f(i) < f(j)$, resp. $f(i) > f(j)$.*

Algorithm 2: Generalised Evolutionary Algorithm

Data: Initial population = pop
Data: N = Number of generations
Result: Optimised population = pop
for $g \in 1..N$ **do**
 $co \leftarrow \text{Crossover}(pop);$
 $m \leftarrow \text{Mutate}(co);$
 $pop \leftarrow \text{Select}(m);$
end
return pop

Let us review the meaning of the aforementioned operators.

- crossover usually involves recombination of features from two or more individuals from the population. It's purpose is to provide means of exploitation of features beneficial to the population already developed by some individual.

- mutation provides means of exploration by introducing randomness into the individuals EA without crossover would be more akin to a random walk algorithm.
- selection then is about filtering the resulting individuals based on the objective function of the algorithm. It might be performed deterministically or with varying degree of stochasticity as elaborated in section 3.1.

It should be noted that the order of mutation and cross-over here are arbitrary and are specific to each particular evolutionary technique. For further discussion of selection process we need to introduce a few terms specific to evolutionary algorithm that have been borrowed from their counterpart in biology.

3.1 Continuous Optimisation

Historically the individuals optimised by EA used be primarily conceived as vectors over binary domain - $\{0, 1\}$. However this approach was later expanded into the domain of vectors over real numbers. In literature this is referred to as continuous optimisation. In this thesis we are interested in neuro-evolution, a subset of continuous optimisation as neural networks discussed in the chapter 1 are represented as a series of vectors of real numbers. But since the intend of this thesis is mainly to compare different objective functions we will not delve too deeply into the intricacies of neuro-evolution field and instead approach the following techniques from the more general standpoint of continuous optimisation.

In this section we will review the selected evolutionary algorithms and describe them as they are going to be used during the experiments in chapter 5.

3.1.1 Evolutionary Strategies

We base information in the section mainly on the [Beyer and Schwefel, 2002] paper. Evolutionary strategies (ES) are very direct incarnation of algorithm 2 into the realm of real numbers.

We recognise two types of evolutionary strategies: $\mu + \lambda$ and μ, λ . μ denotes size of already existing population while λ denotes size of new population pool. In $\mu + \lambda$ selection happens from the existing population pool and the new population pool at the same time while the μ, λ variant only ever selects from the new population pool. This makes the evolution $\lambda + \mu$ more stable as established successful individuals have a bigger chance to pass to the next generation unless a more successful individual has been found. That said, for some problems this approach risks stagnation and it is preferable to use μ, λ . However the experience with our selected environments from chapter 2 clearly showed dominance of $\mu + \lambda$ as even for the Cartpole, simpler of the environments, improving mutations were too costly to be lost on an unlucky new generation.

Therefore all further discussion of evolutionary strategies will be focused on $\mu + \lambda$. We detail the procedural aspects of $\mu + \lambda$ in the algorithm 3.

Mutation

Distinct feature of evolutionary strategies is the mutation operator that takes as a parameter standard deviation σ and which mutates the value stochastically by a number $y \sim N(0, \sigma)$. In the literature we can often find ES where the mutation σ is part of the individual genotype this might help the evolution as more successful individuals might enforce milder mutation while less successful individuals might mutate wildly in search of better genotype. For our purposes this however introduces two problems. Firstly it complicates interpretability of the results and since we are more interested in the comparison than raw results we cannot ignore this downside. Secondly this approach intuitively works well for cases where objective function is also the meta-measure we actually care about but we are precisely interested in comparison with cases where the objective function is rather orthogonal to actual usefulness of the individual as we will discuss in section 3.2. There this approach could be chaotic at best and detrimental at worst.

For these reasons we opt for a global σ that does not change during the run of the evolution.

Crossover Methods

Crossover is a method of combining values from two selected individuals. Based on We will discuss mainly two kinds of cross-over:

- **Arithmetic** crossover where we combine genotype vectors x_1, x_2 into a new genotype x' as follows:

$$x' = \frac{x_1 + x_2}{2}$$

- **Uniform** crossover where iterate over components of two parent genotype vectors x, y through index $i \in 1 \dots d$ where d is the size of the genotype (we assume both parents are the same length). Then with parameter p_u we produce new individual's genotype v like so:

$$v_i = \begin{cases} x_i & \text{if } r \sim \text{Uniform}(0, 1) < p_u \\ y_i & \text{otherwise} \end{cases}$$

We do not consider position based cross-over methods as they might become quite difficult to manage for the neuroevolution purposes.

Tournament selection

Selection in evolutionary algorithms can be realised by various means such as roulette selection where each individual i has likelihood of passing into the next generation p_i as follows: $p_i = \frac{f(i)}{f(pop)}$, where $f(i)$ is objective function value for the individual i and $f(pop)$ is sum of objective value of all individuals. This however works only for objective functions that give out only either negative or positive numbers. As we have described in chapter 2 environments explored in this thesis can give out rewards with value negative or positive based on the

agent's performance. Therefore we will utilise tournament selection described in Goldberg and Deb [1991]. Tournament selection works on simple principle where we select pool of individuals, size of which is called *tournament size*. From them we then select the one with the most optimal objective function value [Goldberg and Deb, 1991]. We detail the workings of tournament selection in algorithm 3. Note that for the purposes of this thesis we have selected *tournament size* = 3, or ternary tournament. Bigger tournament size provides greater stability to the search as the more optimal individual are given more chances to be selected. The ternary tournament has been selected on empirical basis specifically for our environments.

Algorithm 3: Evolutionary Strategy $\mu + \lambda$

Input: Population size μ , new generation pool size λ , crossover rate CR , mutation rate MR , objective function $f(x)$, crossover operator $CrossOp(x)$, mutation power σ , number of generations G

Initialise population $\{x_i^0\}_{i=1}^\mu$;

Evaluate $f(x_i^0)$ for all i ;

// x^0 forms initial population

for $g = 0$ **to** $G - 1$ **do**

for $i = 1$ **to** λ **do**

 // Crossover:

 randomly choose $r_{CR} \sim Uniform(0, 1)$;

if $r_{CR} < CR$ **then**

 Select random individuals $a, b \sim Uniform(1, \mu)$;

$$v_i \leftarrow CrossOp(x_a^g, x_b^g)$$

else

 Select random individual $a \sim Uniform(1, \mu)$;

$$v_i \leftarrow x_a^g$$

 // Mutation:

 randomly choose $r_{MR} \sim Uniform(0, 1)$;

if $r_{MR} < MR$ **then**

$$v'_i = v_i + \sigma \mathcal{N}(0, 1)$$

else

$$v'_i = v_i$$

 Evaluate $f(v'_i)$;

 // v' forms the new population pool

$v'_i \rightarrow x_{\mu+i}^g$;

 // Tournament selection:

for $i = 1$ **to** μ **do**

 Select random individual $a, b, c \sim Uniform(1, \mu + \lambda)$;

$$x_i^{g+1} \leftarrow \arg \max(f(x_a), f(x_b), f(x_c))$$

return x^G

3.1.2 Differential Evolution

In the introductory paper Storn and Price [1997] on which this section is mostly based, the differential evolution is characterised mainly by its industrial capabilities such as easy parallelisation and application.

Because of these benefits it is a popular technique in the field of neuroevolution. We detail the whole process in algorithm 4. The algorithm for resolving a single generation happens in a single loop with the familiar stages which we detail in algorithm 4.

Selection

Unlike in evolutionary strategies described in section 3.1.1, selection is done on individual basis to enhance parallelisation capabilities of the algorithm. This means that for each individual x_i we either select it or its proposed replacement v_i based on the values of objective function for each of them. Of course we keep the more successful one.

Mutation

While Evolutionary strategies introduce stochasticity in explicit and controlled manner, differential evolution relies on a special form of contextual mutation where three individuals from the population are drafted and combined. This means that we do not have to search for optimal value of σ .

We select three individuals x_a, x_b, x_c distinct from x_i and compute a new individual: $x'_i = x_a + \alpha(x_b - x_c)$ where x_a is called anchor, $(x_b - x_c)$ differential and α is scaling factor.

Although the scaling factor α does not have probabilistic interpretation, in some sense it corresponds to mutation rate in ES as the grater it is, the more stochasticity affects the system.

Recombination

During recombination we decide for each vector component of the newly forming individual x'_i whether we will source it from the already mutated version x'_i or the original x_i based on recombination rate γ . Recombination is rough translation of the concept of crossover for the purposes of single easily parallelised loop and recombination rate could be likened to the uniform crossover cross-rate R_U .

3.1.3 Archives

A technique that may enhance the techniques presented in this sections is installing an archive into the algorithm. Archive is a long-term storage outside of the optimised population of a certain size where a batch of individuals selected from regular population is added periodically and are being reintroduced during the selection. This approach can help preserve diversity of individuals or on the contrary to preserve the best solutions found so far. We will distinguish archives intended for the second purpose as *elite archives*.

Algorithm 4: Differential Evolution

Input: Population size N , scaling factor α , recombination probability γ ,
objective function $f(x)$, number of generations G
Initialise population $\{x_i^0\}_{i=1}^N$;
Evaluate $f(x_i^0)$ for all i ;
for $g = 0$ **to** $G - 1$ **do**
 for $i = 1$ **to** N **do**
 // Mutation:
 Select random individuals $a, b, c \neq i$;

$$v_i = x_a^g + \alpha(x_b^g - x_c^g)$$

 // Crossover:

$$u_{i,j} = \begin{cases} v_{i,j} & \text{if } r \sim \text{Uniform}(0, 1) < \gamma \\ x_{i,j}^g & \text{otherwise} \end{cases}$$

 // Selection:
 Evaluate $f(u_i)$;
 if $f(u_i) \geq f(x_i)$ **then**
 | $x_i^{g+1} \leftarrow u_i$;
 else
 | $x_i^{g+1} \leftarrow x_i^g$;
 return x_i^G

For evolutionary strategy we simply add the archive batch from which a new generation pool is being selected as this will not affect the run of the algorithm in any way.

For differential evolution we cannot just add the archive to the existing population. We actually have to reintroduce the archived individuals instead of parts of the population. In this thesis we will do this on basis of random pick from the *population + archive*.

3.2 Objective functions

In this section we will discuss possible objective functions for the evolutionary optimisation search.

3.2.1 Fitness

Using objective function as the utility function of the algorithm is the most direct way of establishing relationship between the solution we want and the individuals in the evolving population.

Though this approach ensures that the most optimal individuals from given population will generally be selected into the next generation during the evolution it does not guarantee that their potential for evolving is not diminished either. This way the evolution using fitness can easily get stuck in local optimum

of the objective function. While in most cases it might be still possible to reach global optimum on the given problem by changing the hyper-parameters to favour random exploration, it may come to point where the benefits of evolutionary approach are made void.

Archiving

As stated in section 3.1.3 for fitness the two main reasons for introducing archive into an fitness based EA is either to promote diversity by randomly preserving part of the population which might have been eradicated to the detriment of the overall evolution or on the contrary to preserve the best performing individuals so they may not be accidentally selected out of the gene pool.

3.2.2 Novelty

In its introductory paper novelty Lehman and Stanley [2011] is conceived as a guide of the evolutionary search to help avoid traps of local optima.

Definition 19 (adapted from [Doncieux et al., 2019]). *Let E be environment and h_t be a history up to a time t and let $d(x, y)$ be a metric over set B . Then (B, d) is a behavioural space if there is a function $\omega_B : h_t \rightarrow B$, called observer function.*

For the purposes of this thesis the observer function is implemented through the $agg(x)$ and $transform(x)$ functions used in algorithm 1.

Definition 20 (adapted from [Lehman and Stanley, 2011]). *Let P be a population of individuals within EA and (B, d) be a behavioural space. Then for each individual a we define novelty:*

$$N(a) = \frac{\sum_{b \in P, b \neq a} d[\omega_B(a), \omega_B(b)]}{|P| - 1}$$

With novelty as objective function the selection process incentivises diversification of behaviour among the population. This allows the evolutionary search to retain variety within the population across generations, leveraging the crossover as well as mutations to leave local optima.

Downside of this approach might be in the ambiguity in the choice of behavioural space. Incorporating too many environment variables will lead to faux-exploration in variables completely orthogonal to fitness. On the other hand leaving out an important variable means the algorithm will not retain diversity where it matters, making novelty pointless. This means the behavioural function and space have to be selected empirically for given problem.

If we do not state otherwise in this thesis novelty will be calculated in relation to the upcoming generation $g + 1$. During the unrecorded preliminary experiments we have not found any significant advantage to other methods of computation.

Archiving

In fitness the random archive had a role of providing variety, which is the main goal of novelty search. Here we actually hope to use the randomness as a stabilising element since reintroduction of old behaviours will prompt the new generation

to avoid them. This approach was with success used in Gomes et al. [2015], outperforming pure novelty and novelty based archive.

To implement elitist version of archive for novelty we will introduce a fitness threshold T_f to the random archive so that given a randomly selected batch Q_{rand} and fitness function f we screen the selected individuals so that we get actual elite batch Q_{elite} :

$$Q_{elite} = \{i \in Q_{rand} | f(i) \geq T_f\}$$

Only these screened individuals are then added to the archive. This will help us explore the idea of the archive's stabilisation role.

3.2.3 Combinations of Novelty

The novelty does not take into account the individual's performance with objective function. Therefore when two individuals with similar behavioural characteristics appear but one of them performs significantly better than the other, pure novelty algorithm can easily select the less performant individual, losing the more performant genome for further generations. To prevent this we want to meaningfully combine novelty with fitness.

Exponential Decay Weight

Taking inspiration from Cuccu and Gomez [2011], we will try to combine novelty with fitness in the most straight-forward way - linear combination with decaying weights. However first we need to address some issues such as the disproportion in scale of fitness to scale of novelty. This mean we require normalisation for the two. Again taking inspiration from Cuccu and Gomez [2011], we will normalise with the maximum and minimum of both fitness and novelty in the currently existing population for each individual i :

$$\hat{f}_i = \frac{f_i - f_{\min}}{f_{\max} - f_{\min}}, \quad \hat{n}_i = \frac{n_i - n_{\min}}{n_{\max} - n_{\min}}$$

Both maximum and minimum are easy to obtain as they are one $O(n)$ while other methods such as z-score might be computationally more demanding in the multiplicative constant. Then we get combination:

$$c_i = W\hat{f}_i + W'\hat{n}_i$$

Unlike Cuccu and Gomez [2011] we will however decay the weight W of fitness or $W' = (1 - W)$ of novelty over time. So that for base W_b , decay rate r_D , current generation g , total number of generations G and **increasing novelty** we have:

$$W = W_b e^{-r_D \frac{g}{G}}$$

Conversely for **increasing fitness**

$$W' = W_b e^{-r_D \frac{g}{G}}$$

This will allow us to explore whether the exploratory stage has more meaning at the beginning when it generates diverse population that can later pruned by the shift for fitness or is it better to first evolve the population to certain level and then explore with more adept individuals.

4. Comparison criteria and conditions

To compare different types of objective functions fairly, we have to select criteria by which we compare them and establish conditions under which will this comparison take place so that our conclusions are fair.

4.1 General criteria

Since fitness provides direct measure of performance we would like to make our comparison on the fitness of the final population and its distribution in it. To that end we have to perform experiments with variety of initialisations of the respective environments, getting more comprehensive picture then from a single run. This will be mainly reflected in the seed for random numbers generator of the environment. Since the promise of the novelty approach lies mainly in avoiding the trap of local optima we also want to review statistics of each generation produced by the algorithm runs

However we also have to take into account the hyper-parameters with which these populations have been achieved. This is especially (but not limited to) population size and number of generations, hyper-parameters responsible for most of the computational complexity of a evolutionary algorithm run.

4.2 Containers

While we have reviewed evolutionary algorithms and proposed different objective functions in chapter 3, we now have to somehow refer to the combination of the two. We use the term **container** to denote a concrete combination of an evolutionary algorithm, objective-handling method. We can divide the containers on the axis of algorithm to ES containers and DE container. And we can divide them on basis of objective function handling:

- **fitness** container implements pure fitness approach as described in section 3.2.1
- **novelty** container implements pure novelty approach as described in section 3.2.2
- **fitness archive** container implements fitness search with random selection archive as described in section 3.2.1
- **elite archive** container implements fitness search with elite selection archive as described in section 3.2.1
- **novelty archive** container implements novelty search with random selection archive as described in section 3.2.2

- **limit archive** container implements novelty search with fitness threshold archive as described in section 3.2.2
- **increase novelty** container implements increasing novelty objective as described in as described in section 3.2.3
- **increase fitness** container implements increasing fitness objective as described in as described in section 3.2.3

Specifically for ES the crossover method is considered a hyperparameter that will be pinpointed during the Bayes search described in section 4.4.1.

4.3 Individual hyper-parameter selection

The hyper-parameters characterising individuals are number of layers and the number of neurons in their layers. The choice of these parameters is specific for each task. Generally the more complex is the task the more of either layers or neurons per layer the handling NN requires to be suitable for it.

4.3.1 NN Structure

As discussed in chapter 1 as a choice we are making our agents to be simple feed-forward neural networks. This choice is on account of simplicity but also on account of adequacy as the bigger the neural network is the more resources need to be spend on optimising it. The NN structures are depicted on figures 4.1 and 4.2 for Cartpole and Lunar Lander respectively.

Both structures have been selected on account of preliminary experience with the environments. Shapes of input and output layer are determined by the respective observation and action space.

The genome of evolving individual consists purely of the weight and bias matrices of the hidden layers nothing else is subject to evolution. For the purposes of this thesis and in relation to the descriptions of evolutionary mechanics given so far we should think about these matrices as flattened to a single long vector. In the case of Lunar lander with multiple layer the vectors are concatenated.

4.3.2 Behavioural Space

As we can see through the definition of observer function in definition 19, the choice of transformation from history to behavioural space can be arbitrarily complex. Recently there have been proposals of automatic behavioural space extraction like in the paper Meyerson et al. [2016] or Salehi et al. [2021]. However the original paper Lehman and Stanley [2011] emphasises application of domain knowledge to the behavioural space design. Because this thesis is in the core comparative and we are interested in the interpretability of the design as much as raw performance, we have opted for manual design based on our preliminary experiments and experience with the environments. We also take into account the fact that the greatest advantage from an automatic behavioural space extraction is obtained mainly in complex environments [Meyerson et al., 2016].

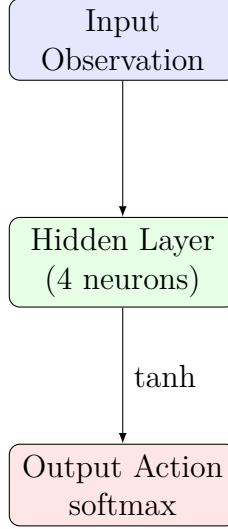


Figure 4.1: Structure of Carpole agentic network

While we won't specify *agg* and *transform* functions introduced in algorithm 1 explicitly for readability purposes, their definitions are implied by respective definition of observer function ω_B and the fact we will only consider the behaviour of the last evaluated episode. While such approach loses some of the information about the evaluated individual it also does not introduce consistency trade-offs of standard aggregations like max, min or *mean* applied component-wise. Their solution is out of the scope of this thesis.

Cartpole

Let $h_{Cart} = (a_0, o_1, a_1, \dots, o_T)$ be history of one completed episode of Cartpole that lasted T steps and o_T is the last observation. Then we define observer function:

$$\omega_B(h_{Cart}) = (pos_X(o_T), \theta(o_T))$$

where $pos_X(o_T)$ is reported position on x-axis in observation o_T and $\theta(o_T)$ is pole angle both normalised to interval $[-1, 1]$ through their respective maximal values reported in table 2.1.

Lunar Lander

Let $h_{Lun} = (a_0, o_1, a_1, \dots, o_T)$ be history of one completed episode of Lunar Lander that lasted T steps. Then we define observer function:

$$\omega_B(h_{Lun}) = (pos_X(o_T), v_Y(\bar{o}))$$

where $pos_X(o_T)$ is reported position on x-axis in observation o_T , v_Y is a reported velocity in y-axis and $\bar{o}$ is mean of all observations recorded in h_{Lun} . Both components of behavioral vector again normalised to interval $[-1, 1]$ through their respective maximal values reported in table 2.3.

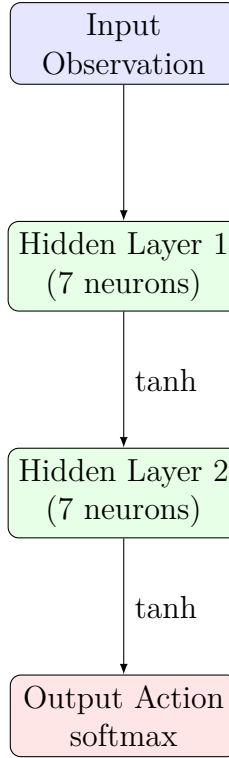


Figure 4.2: Structure of Lunar Lander agentic network

4.4 Algorithm hyperparameter selection

We have reviewed hyperparameters of each algorithm in the chapter 3. Now we will review the process of their selection. The difficulty of the task lies mainly in the complexity of evaluation. As we will discuss later some of the experiments maybe evaluated over day or even several days on the available hardware even for a 100 limited algorithm runs. For this reason we need to utilise bayesian search which is able to prune a lot of the redundant space out so that we may not waste time with it. This however has a considerable downside for our purposes as we would like to compare the algorithms on their best footing. Therefore we refine the bayesian search results with local grid search attempting to deviate from the supposed hyperparameter equilibria.

4.4.1 Bayesian Search

Because we are faced with multiple Bayesian search is an algorithm where we stochastically delineate where a possibly good solution may lie and then try to exploit this delineation. Because the specific workings of Bayesian search are time-consuming to implement ad hoc (and also such implementation is out of the scope of this thesis) We will utilise library Optuna.

Optuna

Just as in our case a lot of machine-learning techniques need for their proper function to obtain adequate hyperparameters first. Optuna is a Python library designed for this purpose [Akiba et al., 2019]. The library itself is multifaceted,

so we will only discuss algorithms and terms pertaining to the subject of this thesis.

Definition 21. *By Optuna trial or just trial we mean one run of the evolutionary algorithm with one chosen set of hyperparameters and the results of this run.*

Definition 22. *Optuna study a set of related trials with implied continuity of choice.*

Definition 23. *Optuna optimisation run describes run of the chosen optimisation algorithm, adding optimised trials to selected study.*

Note that study maybe composed of multiple optimisation runs. In this thesis we have only done one study per container and one optimisation run per study if not for anything than to mind our computational time.

As the optimisation algorithm we have selected Tree-structured Parzen Estimator (TPE) due to the fairly low complexity and effectiveness, wasting no trials as we want to conserve as much actual runtime as possible.

TPE

In this section we will use information from paper Bergstra et al. [2011]. The Tree-structured Parzen Estimator (TPE) used in Optuna is a sequential model-based optimisation method for hyperparameter tuning. It models probability $p(x|y)$ where x is a set of hyperparameters and $y = f(x)$ is observed value of objective function.

It does so by learning two marginal models $l(x)$ a $g(x)$ This split y^* defines what the algorithm currently considers promising based on past experience. The threshold y^* is set to some quantile of observed objective values.

$$p(x|y) = \begin{cases} l(x) & y < y^* \\ g(x) & y \geq y^* \end{cases}$$

Model $l(x)$ represents density learned from observations $\{x_i | f(x_i) < y^*\}$ and model $g(x)$ represents density learned from observations $\{x_i | f(x_i) \geq y^*\}$.

These distributions are typically estimated using smooth non-parametric density estimators (Adaptive Parzen estimators), which can flexibly model hyperparameters without strong assumptions about the shape of the space. The next tried set of hyperparameters is chosen according to the rule:

$$x^* = \arg \max_x \frac{l(x)}{g(x)}.$$

4.4.2 General Gridsearches

Now we will discuss grid searches that are common for all implementations of our EAs. Note that for convenience we do not present these concepts in the temporal order in which they have been evaluated.

Evaluation episodes vs. generations

Before we start other experiments we have to determine how many episodes will be used to evaluate an individual during the algorithm’s run in these experiments. The parameter directly influences how much information the algorithm has extracted about the individual, as more episodes will lead to more indicative behavioural descriptors. However since the episodes take up non-trivial part of the computation we want to select the lowest possible number. For this purpose we will run a grid search comparing differential algorithm’s performance over different number of generations. Since the Gymnasium environments discussed in chapter 2. are based on similar principles we will perform the search only and Cartpole but generalise the results to both. This is done to conserve time for more impactful experiments.

Obtaining the values from the grid search we will select the optimal number of episode based on the following rationale: For each episode count tested we define strategies S where $s \in S$ is the corresponding number of episodes. We name expected performance of strategy s in run of length g as $\mathbb{E}[V_g(s)]$ We define gain of strategy s in run of length g as $A_g(s) = \mathbb{E}[V_g(s)] - \mathbb{E}[V_g]$ where $\mathbb{E}[V_g]$ is the total expected performance in generation g across all strategies. We then select

$$s^* = \arg \max_{s \in S} \left(\frac{\mathbb{E}_{g \sim G}[V_g(s)]}{cost(s)} \right)$$

where $cost(s) = e^s$, introducing exponential cost to strongly penalise increasing episode count, accounting for the overall difficulty associated with the increase.

Generations grid search

During the refining in Bayesian search and Incremental grid search we use fixed number of generations for the optimisations. This mean that we don’t know how many generations are truly best. We perform a grid search over a range of generations. This will also allow us to compare performances and performance curves based on this parameter.

4.4.3 Iterative grid search

Because we obtained the optimised parameters from Bayesian search stochastically we would like to ensure that there is no easily obtainable set of parameters more performant than the set from BS. All however in time consuming fashion. To this end we have devised an algorithm of incremental grid search where we evaluate possibly useful deviations from the supposed local optima. Upon discovering new optima we recentre the search around it. If we are not successful in finding dominating set of parameters we end the search with the last obtained optima. We also end the search after more than 3 re-centring (hops). Since this is only refining algorithm of already functional solution we do not expect long searches.

Algorithm 5: Incremental grid search general

Input: Initial argument set x_0 , number of hops H ,
evaluation algorithm A

Output: Final argument set x

```
 $x \leftarrow x_0;$ 
 $\mathcal{V} \leftarrow \emptyset;$ 
// fitness performances
 $s \leftarrow \emptyset;$ 
// final populations
 $pop \leftarrow \emptyset;$ 
 $s(x_0), diversity(x_0) \leftarrow A(x_0);$ 
for  $h \leftarrow 1$  to  $H$  do
    Generate valid candidate argument sets  $\mathcal{C} \leftarrow \text{Generate}(x, \mathcal{V}, h);$ 
    foreach  $c \in \mathcal{C}$  do
         $s(c), diversity(c) \leftarrow A(c);$ 
        // for  $c \in \mathcal{V}$  we use cached values
         $\mathcal{V} \leftarrow \mathcal{V} \cup \{c\};$ 
    if  $\mathcal{C} \neq \emptyset$  then
         $S^* \leftarrow \max_{c \in \mathcal{C}} s(c);$ 
         $\mathcal{B} \leftarrow \{c \in \mathcal{C} \mid s(c) = S^* \wedge s(x) \leq s(c)\};$ 
        if  $|\mathcal{B}| = 1$  then
             $x \leftarrow \text{only element of } \mathcal{B};$ 
        else if  $|\mathcal{B}| = 0$  then
            // We have not found improving solution
            return  $x, s(x);$ 
        else
             $x \leftarrow \arg \max_{c \in \mathcal{B}} (diversity(c));$ 
return  $x, s(x);$ 
```

The candidates are generated from neighbourhoods of the selected solution x in one or more hyperparameter. For hyperparameter H the neighbourhood is defined by δ_H . We generate $x_{-\delta}$, $x_{+\delta}$, x_0 such that for $x_{-\delta}(H) = x(H) - \delta_H$ and analogically for $x_{+\delta}$. The option of x_0 might not make sense to generate for a single hyperparameter as this is just x repeated but in combination of two or more hyperparameters this ensures we truly check all combinations consisting the neighbourhood of x in those hyperparameters.

4.4.4 $\mu + \lambda$ incremental grid search

For the purposes of $\mu + \lambda$ algorithm we split the Generation and evaluation of candidate solution for the iteration of the main loop into two parts. First part generates candidates in the neighbourhood on the hyperparameters $\mu + CR$, the second $\lambda + MR$. While all hyperparameters interact we claim that these pairs interact very strongly if not strongest as exploitation power of CR is directly interlinked with the stochastic diversity reservoir of the current population. On the other hand the size the new individual pool limits exploration power of mutation rate. It should be noted that the selection process happens after both candidate batch generations and so the candidates in $\lambda + MR$ stage might be generated from

solution already updated by $\mu + CR$ stage. The reason why we do this instead of generating all at once is time conservation and priority based on projected stability of the solution - we claim that alterations to the main population and crossover rate produce more stable algorithm based on preliminary experimentation. If we don't find the solution at first stage algorithm cannot prematurely terminate before the second stage is evaluated negatively as well.

4.4.5 DEs incremental grid search

As we described in section 4.4.4 we split the candidate generation even for the differential evolution. First stage alternates only size of the population. Second stage alternates hyperparameters $\alpha + \gamma$ or the rescale factor and recombination rate introduced in 4. The algorithm again cannot terminate prematurely before both stages have been checked for improving solution.

4.5 Final comparison

In addition to graphical methods suggested in paper Beiranvand et al. [2017] to examine the results of the test we will use several statistical methods. Following advice in the paper García et al. [2009], which is specifically concerned with evolutionary algorithms, we will mainly utilise the non-parametrical tests which we introduce in this section. This enables us to compare results of our algorithm without imposing additional constraints.

The tests represent a null hypotheses H_0 which may or may not be rejected in favour of hypotheses H_1 . This is done based on obtained p-value p representing a probability of observing the obtained data assuming the null hypothesis is true. We define a threshold of significance α so that for $\alpha > p$ the null hypothesis is rejected. Usually values $\alpha = 0.1$ or $\alpha = 0.05$ are chosen.[García et al., 2009]

4.5.1 Friedman test

We will use description of Friedman test from García et al. [2009]. There it is described as a statistical test testing k algorithms. The null hypothesis is that all k algorithm sample from the same population distribution. If rejected we accept at least one of the k algorithms samples from a different population distribution from the rest. We compute rankings r_j^i of each algorithm j for each comparison instance i . In the case of this thesis comparison instance is a initial conditions set for an particular instance of evolutionary algorithm. Rank is distributed from the best having rank 1 to the worst rank k . Then we compute the Friedman statistic χ_F^2 :

$$\chi_F^2 = \frac{12N}{k(k+1)} \left(\sum_{j=1}^k R_j^2 - \frac{k(k+1)^2}{4} \right)$$

where $R_j = \frac{1}{N} \sum_{i=1}^N r_j^i$ and N is the number of comparison instances. The χ_F^2 is then converted to p-value through a combinatorial computation. In reviewed literature it is not common to go over the computational process explicitly as it maybe quite complex in the variety of approximations. We will use the implementation in Python library SciPy which cites Spencer et al. [2025].

4.5.2 Wilcoxon signed-rank test

Introduced in the paper Wilcoxon [1992] in both paired and unpaired version, Wilcoxon signed-rank test enables us to compare two algorithm without assumptions about the distributions of their tested output. We start by computing differences of the algorithms' A and B performances in each instance i of total of N like so:

$$d_i = A_i - B_i \quad (4.1)$$

Unlike in Friedman test where the rank is distribute across algorithms, in Wilcoxon test distributes rank across the absolute value of the differences $|d_i|$. Then we compute sums of ranks testifying for each of the compared algorithm. R^+ is the sum of ranks where algorithm A won ($d_i > 0$) while R^- is the sum of ranks where algorithm B won ($d_i < 0$). For $d_i = 0$ we split the ranks between the two sums [García et al., 2009].

$$R^+ = \sum_{d_i > 0} rank(d_i) + \frac{1}{2} \sum_{d_i = 0} rank(d_i)$$

$$R^- = \sum_{d_i < 0} rank(d_i) + \frac{1}{2} \sum_{d_i = 0} rank(d_i)$$

According to the paper García et al. [2009] we then obtain value

$$T = \min(R^+, R^-)$$

The T is then compared to the value of Wilcoxon distribution with N degrees of freedom. The null hypothesis that both algorithms perform equally is rejected if the T is less or equal. Corresponding p-value is calculated using normal approximation of Wilcoxon T statistic. [Cureton, 1967]

Holm's and Hochberg adjustment

Since we will be performing number of comparisons between different algorithm instances, we need to take measures to prevent family-wise error rate (FWER). To do so García et al. [2009] recommends Holm's and Hochbergs adjustment procedures.

Holm's procedure orders the obtained p-values so that: $p_1 \leq p_2 \leq \dots \leq p_{k-1}$. Each p-value p_i , starting with the most significant, is then compared to $\frac{\alpha}{k-i}$ where α is significance threshold. We continue up the chain as long as we can reject the null hypothesis H_i based on $\frac{\alpha}{k-i}$. We stop once we cannot reject the null hypothesis and we conclude that all subsequent null-hypoteses hold up. [García et al., 2009]

In Hochberg's procedure we start from the least significant p-value and work our way down the chain. For the chain $p_1 \geq p_2 \geq \dots \geq p_{k-1}$ comparing p_i with $\frac{\alpha}{i}$. If the null hypothesis holds up we move on to the next p-value. If it is rejected we consider all subsequent null-hypotheses rejected as well.[García et al., 2009]

Hodges–Lehmann estimator

Hodges-Lehman estimator is method of estimating direction of relationship between two treatments tested by Wilcoxon test published in Jr. and Lehmann [1963]. Taking the differences from eq. (4.1) we calculate the Hodges–Lehmann estimator HL for N differences d_i as such:

$$HL = \text{median} \left(\left\{ \frac{d_i + d_j}{2} \mid 1 \leq i \leq j \leq N \right\} \right)$$

This method should then provide more accurate picture of the comparison than simple median of differences [Jr. and Lehmann, 1963].

5. Experiments

In this chapter we will discuss the experiment and their results. The code for the experiments has been written in Python 3.10 and experiments were conducted on several computers characterised in attachment ???. The complete project with data and scripts is available publicly in Github repository.

5.1 Episode Grid Search

As we have described in the section 4.4.2 we have performed a grid search on the Cartpole environment using the pure fitness method on both evolutionary algorithms (ES and DE).

The generations and episodes over which the search was conducted are

$$g \in \{3, 7, 1015, 20, 30\}$$

$$s \in \{1, 2, 3, 4, 5, 6, 7\}$$

The aggregated results based on the calculation introduced in section 4.4.2 are presented in table 5.1. Based on the results we have selected 3 episodes for internal evaluation within the evolutionary algorithm. For evaluation of produced population at the end of EA's run we will use 5 episodes.

eps	cost adj. gain
1	-8.320725
2	0.153129
3	0.317113
4	0.213275
5	0.078688
6	-0.021315
7	0.000359

Table 5.1: Evaluation episode selection table

5.2 Bayesian Search

As mentioned in section 4.4.1, we perform one optimisation run per container, resulting in one relevant study. This is purely practical measure pertaining to time feasibility of the experiments. In each optimisation run we perform fixed number of trials. We use 150 trials as baseline, adding 10-20 trials for more parametrised containers. Because we are dealing with stochastic method this number does not guarantee neither good or bad performance and has been arbitrarily chosen from experience with preliminary experiments. The hyper-parameter selection for each run is done within limits detailed in table tables 5.2 and 5.3. The hyper-parameters found are presented in tables tables 5.4 to 5.7.

The shorthand arch. T mean period of archiving in generations. W_0 is starting value of W or W' in increasing fitness or increasing novelty respectively. Rate of

Parameter	cartpole			lunarlander		
	Min	Max	Step	Min	Max	Step
method	uniform	arithm.	-	uniform	arithm.	-
μ	10	30	10	40	70	10
λ	10	30	10	40	70	10
mr	0.00	0.10	0.01	0.00	0.50	0.01
cr	0.30	1.00	0.10	0.30	1.00	0.10
σ	0.50	2.00	0.50	0.50	3.00	0.50
arch. T	2	5	1	2	5	1
arch. batch	1	5	1	1	5	1
limit	200	300	10	-200	-50	10
W_0	0.20	0.70	0.10	0.20	1.00	0.10
r_D	0.50	5.00	0.50	0.50	5.00	0.50
p_U	0.10	0.90	0.10	0.10	0.90	0.10

Table 5.2: Hyper-parameters limits for Bayesian search ES

Parameter	cartpole			lunarlander		
	Min	Max	Step	Min	Max	Step
pop	5.00	20.00	5.00	40.00	70.00	10.00
mutation_rate	0.00	1.00	0.10	0.00	1.00	0.10
cross_rate	0.30	1.00	0.10	0.30	1.00	0.10
archiving_period	2.00	5.00	1.00	2.00	5.00	1.00
archive_batch	1.00	5.00	1.00	1.00	5.00	1.00
limit	200.00	300.00	10.00	-200.00	-50.00	10.00
start_fit_w	0.20	1.00	0.10	0.20	1.00	0.10
decay	0.50	5.00	0.50	0.50	5.00	0.50

Table 5.3: Hyperparameters limits for bayesian search DE

decay is labelled r_D . Other parameters should be explained by the notation in chapter 3.

While optuna has internal mechanism for selecting the best trial, sometimes we can obtain number of equivalent trials that are better than all other trials. This concerns mainly Cartpole, where we have on several occasions discovered solving individual for multiple configurations.

When multiple best trials are available we will select the one which has the highest operation rates K where $K = CR + MR$ for ES and $K = \alpha + \gamma$ for DE. The logic behind this is that suppression of exploratory operators leads to greater exploitation of the roll on initial population. Therefore we assume less exploratory algorithm has a higher chance of simply being lucky on the tested random seeds. If the tie is still not broken we select the random trial from the ones with lowest population score $S = \mu + \lambda$ for ES and $S =$ population size for DE.

Particularly interesting are the results of ES where the uniform method of crossover completely dominates both task, despite quite skewed values of parameter p_U .

5.2.1 Iterative grid search

We want to refine the solution obtained during the section 5.2 using our iterative grid search described in section 4.4.3.

To avoid duplication we present the results of the iterative grid search (IGS)

	arch. batch	arch. T	cr	method	p_U	r_D	W_0	μ	limit	λ	mr	σ
fitness	-	-	1.00	U	0.9	-	-	10	-	30	0.09	0.50
pure novelty	-	-	1.00	U	0.3	-	-	10	-	30	0.06	0.50
inc. fitness	-	-	1.00	A	-	0.5	0.3	20	-	20	0.10	1.00
inc. novelty	-	-	1.00	U	0.7	1.0	0.4	10	-	20	0.09	1.00
fit. archive	1.0	5.0	1.00	A	-	-	-	20	-	20	0.09	2.00
elite archive	2.0	3.0	1.00	U	0.5	-	-	20	-	30	0.08	1.50
novetly archive	5.0	5.0	1.00	U	0.4	-	-	20	-	30	0.07	1.50
limit archive	5.0	4.0	1.00	U	0.6	-	-	20	220.0	30	0.06	1.00

Table 5.4: Bayesian Search results Cartpole ES

	arch. batch	arch. T	cr	r_D	W_0	pop	limit	mr
fitness	-	-	1.00	-	-	15.00	-	0.50
pure novelty	-	-	1.00	-	-	20.00	-	0.80
inc. fitness	-	-	0.90	2.5	1.0	10.00	-	1.00
inc. novelty	-	-	1.00	3.5	0.4	10.00	-	0.70
fit. archive	3.0	3.0	0.90	-	-	20.00	-	0.90
elite archive	4.0	2.0	0.50	-	-	20.00	-	0.80
novetly archive	3.0	3.0	0.80	-	-	15.00	-	0.60
limit archive	3.0	3.0	1.00	-	-	20.00	240.0	0.90

Table 5.5: Bayesian Search results Cartpole DE

in delta tables 5.8 to 5.11 derived for hyper-parameter i like so:

$$\delta_i = F_i - O_i$$

where F_i is the found value of hyper-parameter and O_i is the original value passed from step 5.2. This way we can simply add the contents of the delta table to the original to get the final outcome.

We can see that there was a room for improving almost all of the proposed hyper-parameter sets in all of the hyper-parameters subject to refinement.

5.2.2 Generational grid search

Given the refined hyper-parameters obtained in section 5.2.1 we now search for the optimal number of generation for which the evolution should go on. We perform the grid search as described in section 4.4.2.

It should be noted that to obtain the results we have not made a single long algorithm run with sampling of population on the specified generation but every generation count represents it's own generation run. While this strategy is significantly less efficient it is required as some objective handling methods, particularly combinations of novelty and fitness (3.2.3) have aspects that rely on total number of generation.

We have searched over the following generation counts:

- For Cartpole $g \in \{5, 10, 15, 20, 25, 30*, 40*, 50*\}$
- For Lunar Lander $g \in \{10, 15, 25, 35, 50, 60, 80\}$

Because some containers have not been able to reach solving individual in Cartpole in the original scope of the grid search and have exhibited interesting

	arch. batch	arch. T	cr	method	p_U	r_D	W_0	μ	limit	λ	mr	σ
fitness	-	-	1.00	U	-	-	-	40	-	40	0.02	1.50
pure novelty	-	-	0.70	U	0.2	-	-	70	-	40	0.06	2.50
inc. fitness	-	-	1.00	U	0.7	2.0	0.6	70	-	70	0.25	0.50
inc. novelty	-	-	0.50	U	0.7	1.5	0.9	70	-	50	0.28	1.50
fit. archive	4.0	2.0	1.00	U	0.6	-	-	50	-	70	0.01	2.50
elite archive	2.0	4.0	0.90	U	0.8	-	-	70	-	70	0.01	2.50
novetly archive	3.0	5.0	1.00	A	-	-	-	50	-	70	0.10	2.00
limit archive	4.0	5.0	0.50	U	0.5	-	-	70	-90.0	50	0.02	2.50

Table 5.6: Bayesian Search results for Lunar Lander ES

	arch. batch	arch. T	cr	r_D	W_0	pop	limit	mr
fitness	-	-	0.30	-	-	40.00	-	0.06
pure novelty	-	-	0.40	-	-	40.00	-	0.15
inc. fitness	-	-	0.40	4.0	0.5	40.00	-	0.13
inc. novelty	-	-	0.60	1.0	0.4	70.00	-	0.04
fit. archive	2.0	2.0	0.60	-	-	40.00	-	0.30
elite archive	4.0	5.0	0.50	-	-	40.00	-	0.30
novetly archive	5.0	4.0	0.90	-	-	60.00	-	0
limit archive	1.0	2.0	0.80	-	-	50.00	-180.0	0.80

Table 5.7: Bayesian Search results for Lunar Lander DE

enough behaviour we have extended their gridsearches. The extension generation counts have been marked with * symbol.

We showcase the resulting patterns in fig. 5.1 and ??.

Figures 5.1 and 5.2 however have been pruned for readability and do not represent the full range of values that have been taken into account during the ultimate selection such as the population median or mean. The shaded area here represents quantiles q_{25} and q_{75} other values have been pruned. The actual scope of values considered for each contained is represented by chart that have been for readability sake put into section A.1. We evaluated the algorithm on 5 different seeds.

The hand-selected generation counts for the final evaluation are presented in table 5.12.

5.3 Final Results

After obtaining what we believe to be best performing available hyper-parameters for all containers, we apply this to the final round of tests.

We have run the parameterised algorithms on 45 distinct seeds for each environment from which we collected data about the last population. We have subjected the collect data to the statistical tests described in section 4.5. We will now go over the application methods.

First we will test results of both evolutionary algorithms we have utilised (DE and ES) collected across all containers on each of the environments. Then we will group these methods based on objective used, following the internal divide in section 3.2. We will test both internal consistency of the groups and their relationships. Then we will test differences between the objective handling methods granularly. Since the groupings of results are different in each case, we claim

	Δ_{CR}	Δ_{μ}	Δ_{λ}	Δ_{MR}
pure novelty	-0.1	-	-	-
inc. fitness	-	+15.00	-	-
inc. novelty	+0.1	-	-15.00	+0.05
fit. archive	-	+30.00	+15.00	+0.1
elite archive	-0.1	-	+15.00	+0.1
novetly archive	-0.1	+15.00	+15.00	-
limit archive	0.1	-15.00	-	-

Table 5.8: Delta table of hyperparameters for ES on Lunar Lander after IGS step

	Δ_{CR}	Δ_{pop}	Δ_{MR}
pure novelty	+0.1	-10.0	-0.1
inc. fitness	-	+10.0	-
inc. novelty	+0.1	-	-
fit. archive	-0.1	-	-0.1
elite archive	+0.2	+10.0	-
novetly archive	-	-	+0.1
limit archive	-	+10.0	+0.1

Table 5.9: Delta table of hyper-parameters for DE on Lunar Lander environment after IGS step

them to be different test families for the purposes of adjustments to *FWER*.

We will implicitly use the significance threshold $\alpha = 0.05$ as is common in publications.

5.3.1 EA comparison

Since we examine only two distinct EA implementations we have directly performed Wilcoxon test. First we see how the EAs performed on each of the environments separately. On both environments the Wilcoxon test rejects the null hypothesis with p values $p_{LL} = 3.11 \times 10^{-12}$ and $p_{CT} = 0.0002$ for Lunar lander and Cartpole respectively. On each of the environments we calculate the direction of the difference through Hodges–Lehmann estimator (section 4.5.2) over the differences $d_i = ES_i - DE_i$. Directional values are $HL_{LL} = -54.22$ and $HL_{CT} = 29.12$ suggesting that DE has been more performant on Lunar lander while ES has been more performant on Cartpole environment.

We also investigate the diversity in the final populations produced by our algorithms. On Cartpole the difference is not supported with $p_{CT} = 0.49$ but for Lunar Lander environment the difference is supported $p_{LL} < 0.0001$ and skews slightly for the differential evolution $HL_{LL} = -0.122$.

5.3.2 Group comparison

We would like to compare objective methods as such (fitness, novelty, combination of both). To do so we have grouped the objective handling methods according to the division in section 3.2.

	$\Delta\gamma$	Δpop	$\Delta\alpha$
fitness	-	+5.000	-
novelty	0.200	+10.000	-0.100
inc. fitness	-0.100	+10.000	-0.100
inc. novelty	-	+5.000	-
fit. archive	-0.100	-	-0.200
elite archive	-	+5.000	+0.200
novetly archive	-0.100	-5.000	-0.100
limit archive	-0.100	+10.000	-0.100

Table 5.10: Delta table of hyper-parameters for DE on Cartpole environment after IGS step

	ΔCR	$\Delta\mu$	$\Delta\lambda$	Δmr
fitness	-0.100	+5	+5	-
novelty	-	+5	-5	-0.050
inc. novelty	-	+10	-10	-
fit. archive	-0.100	-5	+10	-
elite archive	-0.100	-	-	0.000
novetly archive	-0.100	+5	+5	-0.050
limit archive	-	+5	-	-0.050

Table 5.11: Delta table of hyper-parameters for ES on Cartpole environment after IGS step

Internal Differences

For that we want to assess the group internal differences by the means of Friedman test for groups of 3 and Wilcoxon for the group of 2. We do this separately for each environment. The results are presented in the tables 5.13 and 5.14. p_{DE} denotes p-value for test only on differential evolution runs. Analogically with p_{ES} . Testing the aggregate of both algorithms we get p_{Total} .

P-values rejecting the null hypotheses are highlighted. We can see that in Cartpole differential evolution the combinations of fitness and novelty are both behaving so similarly, we do not have enough evidence to reject assumption that they might be behaving the same. This is even more pronounced in Lunar Lander but only in terms of produced diversity. The fitness group does not produce enough evidence of internal differences only in Lunar Lander and only in aggregate. It should be noted that while we are using academic standard $\alpha = 0.05$ it is not uncommon to use lower standard $\alpha = 0.1$ which would allow us to reject H_0 in several of the aforementioned cases.

Intergroup differences

Having assessed how the groups behave internally, we now compare the groups between each other for each of the environments with Wilcoxon test. When pairing the performance we pair median value of the values achieved by the group members on particular initial seed using a particular algorithm.

The results are presented in tables 5.15 and 5.16 for Lunar Lander environment and in tables 5.17 and 5.18 for Cartpole environment.

Performance wise the fitness group dominates both environments. We can see that combinations dominate over novelty group. In diversity we see that on the

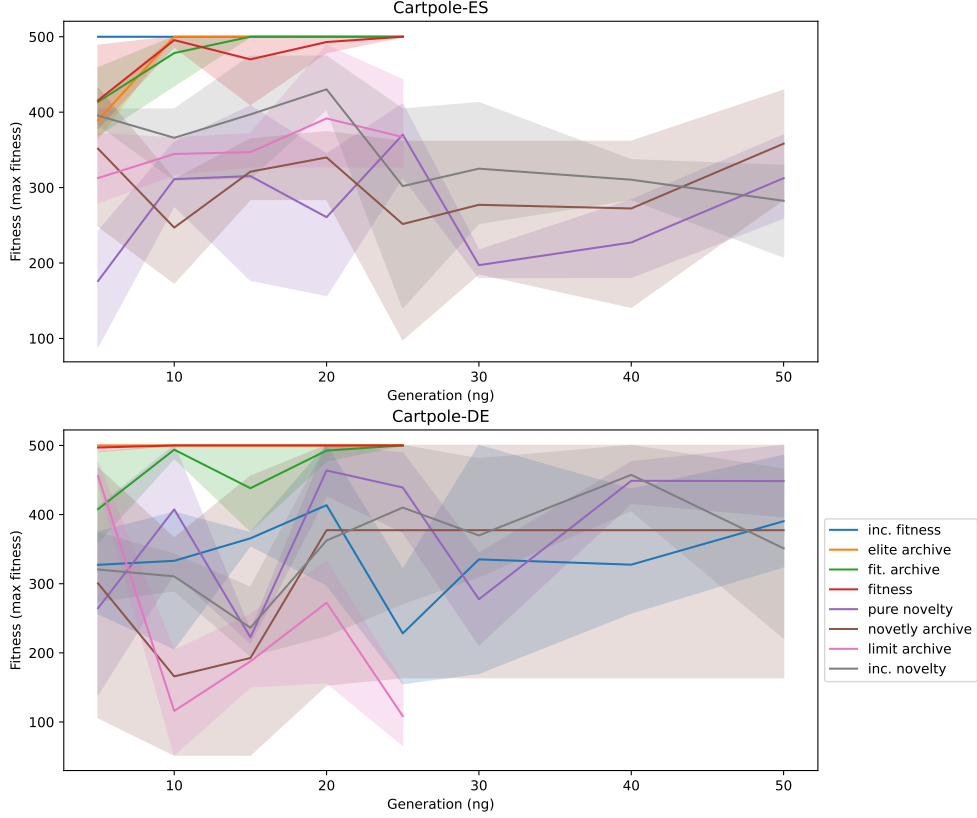


Figure 5.1: Fitness/Generation Characteristics for Cartpole environment (ES up, DE down)

level of individual EAs the differences are statistically significant but in aggregate test the differences may cancel out.

5.3.3 Objective handling comparison

Now we want to compare the objective handling methods based on fitness of individuals they have produced. Friedman test rejects the null hypothesis for both Lunar Lander ($\chi_F = 328.39$, $p_F = 5.1 \times 10^{-67}$) and Cartpole ($\chi_F = 326.66$, $p_F = 1.21 \times 10^{-66}$). So we proceed to the pairwise Wilcoxon tests on both environments to obtain significance of their differences.

The full results for fitness are shown in tables 5.21 and 5.22 respectively. For the cases where the difference wasn't proven significant we do not report the calculated direction. We summarise the results of the comparisons in tables 5.19 and 5.20. The results are summarised in format:

$$\# \text{of won comparisons} / \# \text{of lost comparisons} / \# \text{of ties}$$

. We count tie as either statistically insignificant difference or significant difference with value of Hodges-Lehman estimator $HL = 0$. (section 4.5.2)

We can see that for both environments the pure fitness method dominates with fitness archives being roughly on the same footing. Interestingly pure novelty mostly dominates it's own archival variants.

Elite archive does not show strong dominance over random fitness archive. Even in Cartpole where the null hypotheses of their relationship was rejected,

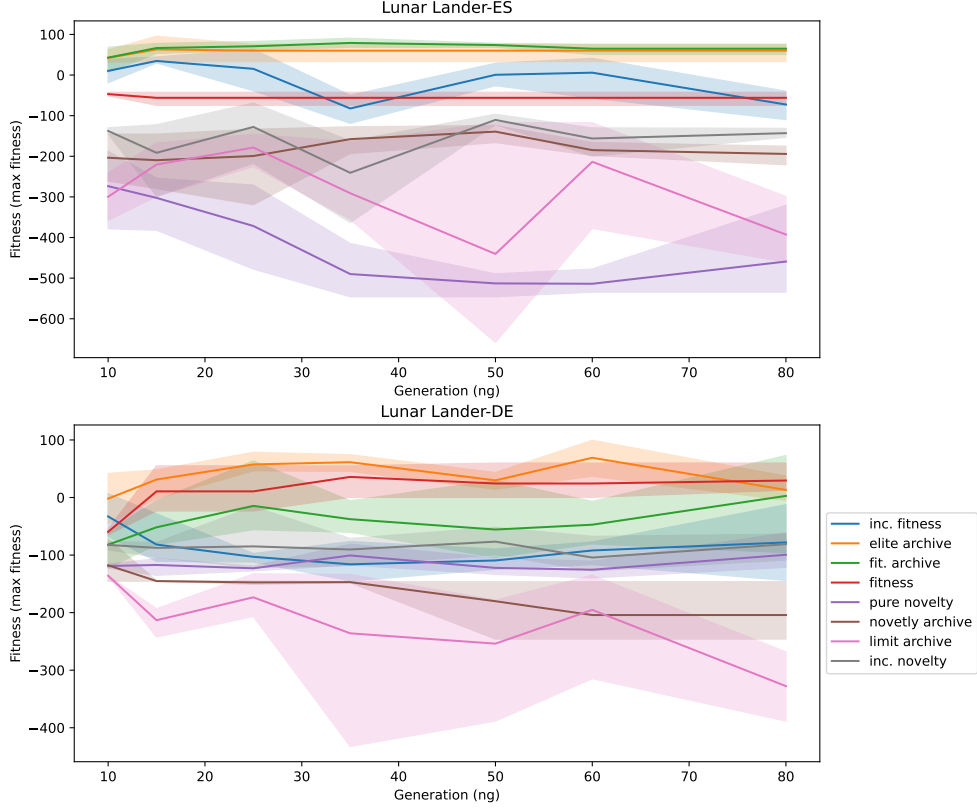


Figure 5.2: Fitness/Generation Characteristics for Lunar Lander environment (ES up, DE down)

the direction came out 0 suggesting the difference to be purely about distribution not systematic shift of the distribution.

Novelty archive dominates limit archive in Lunar Lander environment but in Cartpole limit archive dominates novelty archive.

Diversity Comparison

Now that we have made the comparison on achieved fitness we want to compare the algorithms also based on the produced diversity. The Friedman test rejects the null hypothesis also for diversities both for Lunar Lander ($\chi_F = 192.23$, $p_F = 5 \times 10^{-38}$) and Cartpole ($\chi_F = 163.918$, $p_F = 4.8 \times 10^{-32}$) and so we can safely perform Wilcoxon test on the combinations of objective handling methods. Results are shown in tables 5.23 and 5.24.

5.4 Discussion

We have 16 container identified by the combination of objective handling method and EA used. For all of these 16 containers we have found and refined a set of hyper-parameters used in the final set of experiments. Based on these final experiments we can assess the differences between EAs used, objective handling methods and their groupings.

	Lunar Lander		Cartpole	
	DE	ES	DE	ES
fitness	35	25	15	10
pure novelty	15	10	25	40
fit. archive	80	15	35	25
inc. fitness	15	15	15	20
inc. novelty	25	25	20	40
novetly archive	25	60	20	20
elite archive	60	15	20	10
limit archive	25	25	20	20

Table 5.12: Selection of generations hyper-parameter based on performed grid search

group group name	test test	lunarlander			cartpole		
		$PTotal$	PDE	PES	$PTotal$	PDE	PES
novelty obj.	friedman	6.192×10^{-9}	7.860×10^{-11}	2.569×10^{-4}	0.012	1.424×10^{-8}	8.648×10^{-5}
fitness obj.	friedman	0.096	0.008	0.022	0.030	0.045	0.019
combination obj.	wilcoxon	6.401×10^{-5}	0.009	4.944×10^{-10}	2.739×10^{-6}	0.550	8.402×10^{-8}

Table 5.13: Fitness based tests of group difference

5.4.1 Algorithms

We have found out that each EA dominates different task in performance. This is not completely unexpected as the ES in our case is quite limited with the static sigma and therefore cannot adjust to the needs of the population during it's run on more complex task (Lunar Lander). On the other hand we get this adjustment automatically with DE algorithm without specifying any hyper-parameter. From the performance characteristics in figs. 5.1 and 5.2 we can see that the performance difference may be mostly attributable to the spread of results rather than performance at the top.

This is supported by the fact that in Lunar Lander task the differential evolution clearly produced more diverse populations than ES.

5.4.2 Objective functions

As stated before the fitness objective handling methods dominate the tasks. From tables 5.13 and 5.14 we can also see that their results are also generally more homogenous than the other groups (evidenced by mostly greater p-values).

Although this is true for both environments we claim that in each the causes of the observation are different: in Cartpole the fitness based algorithms reach solving individual and therefore go through a similar process of distribution of this result into the population. On the other hand in Lunar Lander we have unfortunately not been able to achieve solving individual using any of the algorithms or methods. However from the figs. 5.1 and 5.2 we can see that the maximum achieved fitness tends to flatline between well-behaved and very well behaved solution with pure fitness dropping even below this threshold. This indicates presence of false local optima our algorithms have been getting stuck on. This is in line with our idea that Lunar Lander is more complex environment than Cartpole and the driving idea of novelty search [Lehman and Stanley, 2011].

However we do not observe significant improvement in performance for novelty

group group name	test test	lunarlander			cartpole		
		p_{Total}	p_{DE}	p_{ES}	p_{Total}	p_{DE}	p_{ES}
novelty obj.	friedman	7.385×10^{-17}	1.928×10^{-9}	3.454×10^{-11}	6.745×10^{-16}	2.723×10^{-13}	6.773×10^{-5}
fitness obj.	friedman	0.036	1.004×10^{-10}	2.398×10^{-4}	1.802×10^{-13}	6.938×10^{-10}	1.935×10^{-6}
combination obj.	wilcoxon	0.101	0.712	0.088	6.909×10^{-6}	0.082	5.333×10^{-6}

Table 5.14: Diversity based tests of group difference

A	B	HL_{Total}	p_{Total}	HL_{DE}	p_{DE}	HL_{ES}	p_{ES}
novelty obj.	fitness obj.	-175.50	5.410×10^{-16}	-151.06	1.705×10^{-13}	-207.97	3.411×10^{-13}
novelty obj.	combination obj.	-93.14	6.634×10^{-13}	-45.94	3.262×10^{-10}	-147.10	3.173×10^{-9}
fitness obj.	combination obj.	80.45	5.251×10^{-12}	94.26	5.684×10^{-13}	62.18	1.187×10^{-4}

Table 5.15: Fitness based Wilcoxon tests of between objective groups on Lunar Lander environment

A	B	HL_{Total}	p_{Total}	HL_{DE}	p_{DE}	HL_{ES}	p_{ES}
novelty obj.	fitness obj.	0.09061	1.744×10^{-11}	0.05	1.471×10^{-4}	0.14	1.091×10^{-10}
novelty obj.	combination obj.	-	0.071	-0.06	2.306×10^{-7}	0.03	0.036
fitness obj.	combination obj.	-0.101976	3.043×10^{-15}	-0.10	1.705×10^{-13}	-0.10	6.780×10^{-8}

Table 5.16: Diversity based Wilcoxon tests of between objective groups on Lunar Lander environment

based methods nor for methods combining fitness with novelty. The biggest improvement is provided by increasing fitness method in ES algorithm (table 5.21) where it provides more reliable results (fig. 5.3). We propose that in this instance the combination of fitness with novelty does help break out of false optima.

Archiving is of the most benefit to the fitness methods in ES algorithm in Lunar Lander environment where it improved on the pure fitness performance. The elitist and random solution are barely distinguishable with Wilcoxon null hypothesis being often sustained in the comparison or the direction being of negligible magnitude. On the other hand pure novelty almost exclusively dominates it's archival variants both in raw performance and diversity. This suggests that when archives in novelty do provide a benefit they do so by stabilising the population rather than adding more variety to it. However this is unfortunately not the case for our setup.

Limit archive behaves poorly in Lunar Lander but better than random novelty archive in Cartpole, we therefore propose that the difference lies in how difficult is to reach the respective limits within the environment. Since the limits were optimised for it suggests that simply choosing lower limits does not significantly improve on the performance as lesser quality individuals will now be periodically reinserted back into the population. Seeing the relative success of increasing fitness method we propose shifting threshold would be a better solution.

A	B	HL_{Total}	p_{Total}	HL_{DE}	p_{DE}	HL_{ES}	p_{ES}
novelty obj.	fitness obj.	-240.58	1.119×10^{-15}	-294.50	3.359×10^{-8}	-195.25	1.705×10^{-13}
novelty obj.	combination obj.	-91.13	3.697×10^{-7}	-103.54	9.014×10^{-4}	-73.12	1.041×10^{-5}
fitness obj.	combination obj.	150.33	9.573×10^{-15}	185.12	3.359×10^{-8}	114.88	1.934×10^{-7}

Table 5.17: Fitness based Wilcoxon tests of between objective groups on Cartpole environment

A	B	HL_{Total}	p_{Total}	HL_{DE}	p_{DE}	HL_{ES}	p_{ES}
novelty obj.	fitness obj.	-	0.758	-0.12	0.002	0.08	0.001
novelty obj.	combination obj.	-	0.359	-0.25	8.622×10^{-10}	0.15	3.505×10^{-6}
fitness obj.	combination obj.	-	0.107	-0.12	5.479×10^{-10}	0.07	0.006

Table 5.18: Diversity based Wilcoxon tests of between objective groups on Cartpole environment

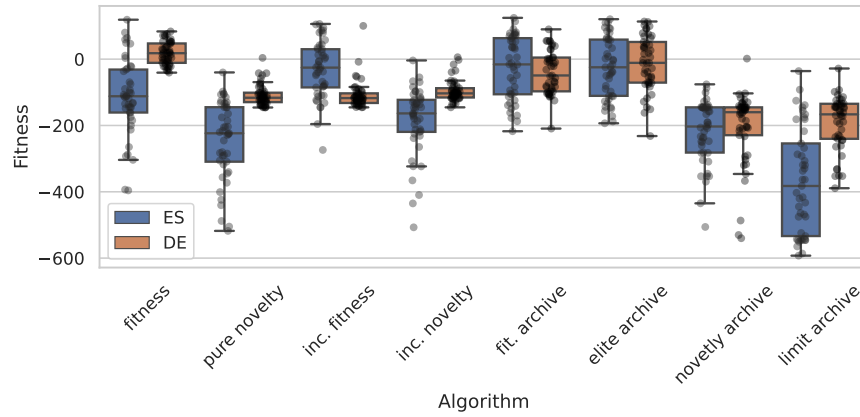


Figure 5.3: Plot of results of final experiment in Lunar Lander environment.

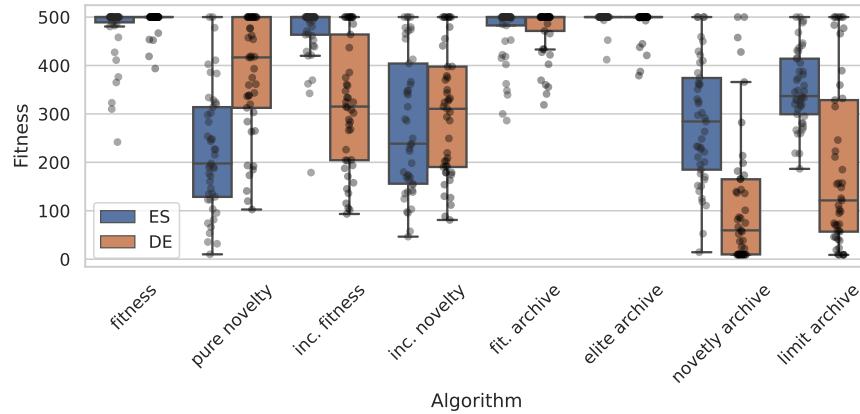


Figure 5.4: Plot of results of final experiment in Cartpole environment.

	Fit. Sum.	Fit. DE	Fit. ES	Div. Sum.	Div. DE	Div. ES
fit. archive	5/0/2	5/1/1	5/0/2	0/4/3	1/4/2	0/5/2
elite archive	5/0/2	5/0/2	5/0/2	0/5/2	0/7/0	1/3/3
fitness	4/0/3	6/0/1	3/3/1	1/4/2	3/3/1	0/4/3
inc. fitness	4/2/1	2/3/2	5/0/2	4/1/2	5/0/2	3/2/2
inc. novelty	3/4/0	2/3/2	1/3/3	4/0/3	5/0/2	4/1/2
pure novelty	2/5/0	2/3/2	1/4/2	6/0/1	5/0/2	5/0/2
novetly archive	1/6/0	0/6/1	1/4/2	4/1/2	1/3/3	6/0/1
limit archive	0/7/0	0/6/1	0/7/0	0/4/3	1/4/2	0/4/3

Table 5.19: Wilcoxon comparison summary fo Lunar Lander env.

	Fit. Sum.	Fit. DE	Fit. ES	Div. Sum.	Div. DE	Div. ES
fitness	5/0/2	5/0/2	4/1/2	5/1/1	4/1/2	2/1/4
fit. archive	5/0/2	5/0/2	4/1/2	2/3/2	2/4/1	2/1/4
elite archive	5/0/2	5/0/2	7/0/0	0/5/2	1/4/2	0/6/1
inc. fitness	4/3/0	2/3/2	4/1/2	2/3/2	4/0/3	0/6/1
pure novelty	1/4/2	3/3/1	0/5/2	6/0/1	4/0/3	6/0/1
inc. novelty	1/4/2	2/4/1	0/5/2	5/0/2	5/0/2	2/0/5
limit archive	1/4/2	0/6/1	3/4/0	0/3/4	1/5/1	2/1/4
novetly archive	0/7/0	0/6/1	0/5/2	0/5/2	0/7/0	2/1/4

Table 5.20: Wilcoxon comparison summary fo Lunar Lander env.

A	B	HL_{Total}	PT_{Total}	HL_{DE}	P_{DE}	HL_{ES}	P_{ES}
fitness	pure novelty	131.676947	2.004×10^{-11}	127.795874	3.070×10^{-12}	134.345431	1.119×10^{-4}
fitness	inc. fitness	-	0.240	128.532971	6.094×10^{-10}	-74.627906	0.005
fitness	inc. novelty	93.049699	4.302×10^{-9}	114.128641	3.070×10^{-12}	-	0.069
fitness	fit. archive	-	0.861	54.150964	4.266×10^{-5}	-83.616144	0.011
fitness	elite archive	-	0.360	-	0.231	-88.211462	0.007
fitness	novetly archive	168.466936	1.409×10^{-11}	207.896223	3.070×10^{-12}	110.273594	6.528×10^{-4}
fitness	limit archive	235.629767	2.223×10^{-13}	204.199313	1.592×10^{-12}	268.610349	2.856×10^{-8}
pure novelty	inc. fitness	-92.98606	5.518×10^{-7}	-	1.000	-208.065336	4.188×10^{-10}
pure novelty	inc. novelty	-28.248713	0.012	-	0.231	-	0.075
pure novelty	fit. archive	-128.184049	8.741×10^{-12}	-70.965488	8.783×10^{-7}	-216.946715	9.053×10^{-9}
pure novelty	elite archive	-146.481877	1.063×10^{-12}	-100.578474	4.508×10^{-7}	-212.219335	9.962×10^{-10}
pure novelty	novetly archive	36.000119	0.015	77.254385	2.242×10^{-8}	-	1.000
pure novelty	limit archive	97.498708	5.715×10^{-8}	77.441572	1.977×10^{-7}	125.560413	6.528×10^{-4}
inc. fitness	inc. novelty	54.464739	6.401×10^{-4}	-	0.053	150.662294	9.393×10^{-9}
inc. fitness	fit. archive	-43.212918	6.486×10^{-4}	-69.290076	1.615×10^{-6}	-	1.000
inc. fitness	elite archive	-51.222516	0.001	-97.50726	1.080×10^{-7}	-	1.000
inc. fitness	novetly archive	138.320717	7.926×10^{-13}	79.183149	1.080×10^{-7}	186.831027	1.577×10^{-9}
inc. fitness	limit archive	200.763637	3.655×10^{-14}	77.426754	1.960×10^{-8}	338.863942	1.295×10^{-7}
inc. novelty	fit. archive	-96.897141	9.969×10^{-10}	-54.684347	4.266×10^{-5}	-153.997883	4.941×10^{-7}
inc. novelty	elite archive	-111.446886	4.632×10^{-12}	-83.858434	5.516×10^{-7}	-147.932789	9.053×10^{-9}
inc. novelty	novetly archive	67.581635	2.382×10^{-6}	89.305449	9.393×10^{-9}	-	0.201
inc. novelty	limit archive	132.155841	1.435×10^{-10}	88.211582	2.102×10^{-8}	189.355484	5.771×10^{-6}
fit. archive	elite archive	-	0.582	-	0.231	-	1.000
fit. archive	novetly archive	174.285356	3.655×10^{-14}	156.265235	1.376×10^{-10}	195.705804	3.595×10^{-10}
fit. archive	limit archive	242.093519	1.409×10^{-14}	151.852576	2.017×10^{-10}	348.97078	7.958×10^{-12}
elite archive	novetly archive	181.016289	8.320×10^{-15}	179.845674	9.152×10^{-12}	182.623177	1.478×10^{-11}
elite archive	limit archive	253.778388	6.829×10^{-15}	174.962109	3.070×10^{-12}	348.347918	1.074×10^{-11}
novetly archive	limit archive	62.807367	0.011	-	1.000	153.728544	1.119×10^{-4}

Table 5.21: Fitness Wilcoxon test over objectives handlers for Lunar Lander

A	B	HL_{Total}	p_{Total}	HL_{DE}	p_{DE}	HL_{ES}	p_{ES}
fitness	pure novelty	179.666667	6.542×10^{-12}	100.75	3.868×10^{-5}	260.333333	5.716×10^{-7}
fitness	inc. fitness	79.916667	2.858×10^{-6}	171.666667	4.424×10^{-6}	-	1.000
fitness	inc. novelty	195.416667	8.411×10^{-13}	189.75	1.471×10^{-6}	198.833333	1.869×10^{-6}
fitness	fit. archive	-	0.444	-	0.141	-	1.000
fitness	elite archive	-	0.438	-	0.960	-0.5	0.045
fitness	novetly archive	287.5	4.660×10^{-14}	410.416667	2.893×10^{-7}	190.666667	6.616×10^{-7}
fitness	limit archive	212.666667	4.471×10^{-13}	316.583333	6.657×10^{-7}	126.75	3.532×10^{-6}
pure novelty	inc. fitness	-97.25	0.002	-	0.138	-253.833333	3.301×10^{-7}
pure novelty	inc. novelty	-	0.687	74.75	0.002	-	0.285
pure novelty	fit. archive	-168.083333	3.433×10^{-11}	-80.916667	0.002	-252.25	3.136×10^{-7}
pure novelty	elite archive	-190.083333	2.480×10^{-12}	-100.083333	5.975×10^{-5}	-279.5	3.136×10^{-7}
pure novelty	novetly archive	108.416667	0.003	274.416667	2.007×10^{-9}	-	0.285
pure novelty	limit archive	-	0.687	200.833333	2.433×10^{-5}	-130.583333	1.113×10^{-5}
inc. fitness	inc. novelty	110.5	3.287×10^{-5}	-	0.960	191.5	1.344×10^{-6}
inc. fitness	fit. archive	-63.416667	9.251×10^{-5}	-155.416667	2.570×10^{-5}	-	1.000
inc. fitness	elite archive	-86.083333	4.789×10^{-9}	-171.083333	3.237×10^{-6}	-9.75	0.002
inc. fitness	novetly archive	200.333333	1.522×10^{-12}	217.25	2.169×10^{-7}	188.333333	8.534×10^{-7}
inc. fitness	limit archive	128.083333	3.061×10^{-8}	136.166667	0.002	121.666667	5.313×10^{-6}
inc. novelty	fit. archive	-182.5	1.214×10^{-12}	-168.5	2.299×10^{-6}	-195.583333	1.237×10^{-6}
inc. novelty	elite archive	-204.333333	3.442×10^{-13}	-187.833333	1.630×10^{-6}	-217.666667	5.577×10^{-7}
inc. novelty	novetly archive	87.416667	0.003	182.833333	3.497×10^{-7}	-	1.000
inc. novelty	limit archive	-	0.687	121.666667	0.021	-78.75	0.023
fit. archive	elite archive	0.0	0.006	-	0.230	-1.416667	0.034
fit. archive	novetly archive	276.083333	5.750×10^{-14}	383.75	2.893×10^{-7}	187.416667	8.534×10^{-7}
fit. archive	limit archive	198.5	3.545×10^{-13}	298.583333	7.852×10^{-7}	117.25	6.848×10^{-7}
elite archive	novetly archive	302.833333	3.903×10^{-14}	403.583333	2.893×10^{-7}	215.666667	5.281×10^{-7}
elite archive	limit archive	220.833333	7.593×10^{-14}	315.916667	7.102×10^{-7}	145.416667	3.136×10^{-7}
novetly archive	limit archive	-68.416667	0.003	-	0.138	-71.75	0.036

Table 5.22: Fitness Wilcoxon test over objectives handlers for Cartpole

A	B	HL_{Total}	p_{Total}	HL_{DE}	p_{DE}	HL_{ES}	p_{ES}
fitness	pure novelty	-0.042555	0.036	-	1.000	-0.100771	0.018
fitness	inc. fitness	0.090809	1.331×10^{-4}	-	1.000	0.198355	2.773×10^{-7}
fitness	inc. novelty	-	0.505	-0.04303	0.046	-	1.000
fitness	fit. archive	0.073376	0.006	0.149382	1.206×10^{-5}	-	1.000
fitness	elite archive	0.217708	2.037×10^{-12}	0.228213	2.306×10^{-10}	0.200677	1.753×10^{-6}
fitness	novetly archive	0.225698	3.939×10^{-7}	0.460536	1.251×10^{-10}	-	1.000
fitness	limit archive	0.146048	1.700×10^{-5}	0.281818	2.306×10^{-10}	-	1.000
pure novelty	inc. fitness	0.149859	2.568×10^{-6}	-	1.000	0.317542	1.994×10^{-8}
pure novelty	inc. novelty	-	1.000	-	0.182	-	0.351
pure novelty	fit. archive	0.129778	4.305×10^{-9}	0.144914	2.803×10^{-7}	0.1164	5.134×10^{-4}
pure novelty	elite archive	0.262882	2.182×10^{-14}	0.234933	2.916×10^{-11}	0.293846	4.027×10^{-10}
pure novelty	novetly archive	0.272268	2.288×10^{-11}	0.456023	3.695×10^{-11}	0.106029	0.002
pure novelty	limit archive	0.19342	6.699×10^{-10}	0.28676	2.589×10^{-10}	0.110312	0.007
inc. fitness	inc. novelty	-0.114747	1.036×10^{-4}	-	0.411	-0.226368	1.013×10^{-4}
inc. fitness	fit. archive	-	0.982	0.134113	2.369×10^{-5}	-0.193233	9.521×10^{-8}
inc. fitness	elite archive	0.122055	1.544×10^{-4}	0.223159	8.230×10^{-10}	-	1.000
inc. fitness	novetly archive	0.125952	0.047	0.44541	3.665×10^{-10}	-0.18049	9.294×10^{-5}
inc. fitness	limit archive	-	0.950	0.28253	6.380×10^{-9}	-0.196636	3.226×10^{-5}
inc. novelty	fit. archive	0.111492	9.680×10^{-5}	0.173304	9.250×10^{-10}	-	1.000
inc. novelty	elite archive	0.243282	3.123×10^{-11}	0.259667	3.665×10^{-10}	0.222153	5.161×10^{-5}
inc. novelty	novetly archive	0.240971	1.146×10^{-7}	0.493238	2.228×10^{-11}	-	1.000
inc. novelty	limit archive	0.171545	3.194×10^{-5}	0.320165	6.380×10^{-9}	-	1.000
fit. archive	elite archive	0.137525	1.653×10^{-5}	-	0.182	0.188703	1.043×10^{-5}
fit. archive	novetly archive	0.141326	1.912×10^{-4}	0.287077	7.464×10^{-6}	-	1.000
fit. archive	limit archive	-	0.169	0.130784	0.044	-	1.000
elite archive	novetly archive	-	1.000	0.212562	0.001	-0.176027	6.507×10^{-4}
elite archive	limit archive	-	0.160	-	0.739	-0.181613	1.100×10^{-4}
novetly archive	limit archive	-	0.125	-0.133493	0.015	-	1.000

Table 5.23: Diversity Wilcoxon test over objectives handlers for Cartpole

A	B	HL_{Total}	p_{Total}	HL_{DE}	p_{DE}	HL_{ES}	p_{ES}
fitness	pure novelty	-0.108583	8.532×10^{-14}	-0.06247	3.308×10^{-10}	-0.182832	3.595×10^{-10}
fitness	inc. fitness	-0.065009	2.023×10^{-11}	-0.063492	2.700×10^{-11}	-0.072573	1.552×10^{-4}
fitness	inc. novelty	-0.078057	7.540×10^{-12}	-0.064042	1.869×10^{-10}	-0.11046	2.725×10^{-5}
fitness	fit. archive	-	0.109	0.040538	0.014	-	1.000
fitness	elite archive	0.049615	0.015	0.133014	4.640×10^{-10}	-	0.071
fitness	novetly archive	-0.084113	4.679×10^{-5}	-	1.000	-0.19034	8.441×10^{-11}
fitness	limit archive	-	0.585	0.052496	0.002	-	0.442
pure novelty	inc. fitness	0.037016	0.018	-	1.000	0.105617	4.264×10^{-4}
pure novelty	inc. novelty	-	0.298	-	1.000	-	0.096
pure novelty	fit. archive	0.142925	1.829×10^{-13}	0.102334	2.102×10^{-8}	0.188583	7.008×10^{-10}
pure novelty	elite archive	0.175722	1.346×10^{-12}	0.192629	1.592×10^{-12}	0.144022	4.114×10^{-5}
pure novelty	novetly archive	0.043506	0.018	0.08317	1.792×10^{-6}	-	1.000
pure novelty	limit archive	0.13087	6.004×10^{-14}	0.110398	1.215×10^{-9}	0.156843	1.301×10^{-10}
inc. fitness	inc. novelty	-	0.585	-	1.000	-	0.442
inc. fitness	fit. archive	0.090774	1.923×10^{-11}	0.10164	3.470×10^{-9}	0.080122	4.999×10^{-5}
inc. fitness	elite archive	0.120897	3.586×10^{-9}	0.195151	2.956×10^{-12}	-	0.153
inc. fitness	novetly archive	-	0.940	0.087288	4.616×10^{-6}	-0.10732	1.423×10^{-5}
inc. fitness	limit archive	0.085673	1.578×10^{-8}	0.11279	1.165×10^{-8}	0.056607	0.017
inc. novelty	fit. archive	0.109141	7.540×10^{-12}	0.105241	7.640×10^{-10}	0.11848	2.093×10^{-5}
inc. novelty	elite archive	0.140465	4.134×10^{-11}	0.194627	1.592×10^{-12}	0.072567	0.011
inc. novelty	novetly archive	-	0.940	0.085309	2.281×10^{-7}	-0.06723	0.011
inc. novelty	limit archive	0.107013	4.437×10^{-9}	0.118202	2.539×10^{-8}	0.093966	0.003
fit. archive	elite archive	-	0.585	0.092345	2.514×10^{-5}	-0.035036	0.008
fit. archive	novetly archive	-0.107116	3.948×10^{-8}	-	1.000	-0.190565	7.958×10^{-12}
fit. archive	limit archive	-	0.940	-	1.000	-	0.104
elite archive	novetly archive	-0.126178	2.357×10^{-11}	-0.109312	9.317×10^{-7}	-0.144965	2.440×10^{-7}
elite archive	limit archive	-	0.298	-0.078102	2.093×10^{-4}	-	0.442
novetly archive	limit archive	0.093258	1.459×10^{-7}	-	1.000	0.16651	4.188×10^{-10}

Table 5.24: Diversity Wilcoxon test over objectives handlers for Lunar Lander

Conclusion

I have successfully developed and implemented framework for testing evolutionary strategies and differential evolution in several setups, such as fitness, novelty and their respective archiving and a combination of them. I have also proposed behavioural models for both environments Lunar Lander and Cartpole. These behavioural models are a modest success, while Lunar Lander model, unlike Cartpole, does not consistently achieve solution grade score it can help achieve well behaved solution in combination with increasing fitness method.

We have also proposed a hyper-parameter searching algorithm based on Bayesian search. While the search itself is rather limited, this reflects the conditions of the used environments, which are time-exhausting to simulate in large numbers. The algorithm allows us to compare the EAs on their performance given hyper-parameters found by this search.

The areas of comparisons were: absolute performance, rate of convergence and diversity of the final populations. Concerning diversity the novelty based approaches have produced equally or more diverse population however in raw performance we have confirmed that both tasks were dominated by the fitness utilising methods. Considering we have not selected particularly deceptive tasks (although Lunar Lander in our configuration has deceptive optima), this in line with literature delineating the usefulness of novelty such as Lehman and Stanley [2011] or Cuccu and Gomez [2011].

However in case of Lunar Lander the pure fitness has gotten stuck in false optima. While there are more possible explanations for the phenomenon, we have seen partial alleviation the problem by utilising dynamic combination of fitness and novelty as the objective. But in our case it ultimately did not out-perform fitness based archiving methods.

Despite, the general approach shows greater promise as we could expect that a more powerful behavioural characteristic might be discovered using automatic methods that were outside of the scope of this thesis. On the other hand there is a limited room to refine the fitness based archive substantially over the version presented in this text without altering its core concept.

We have also confirmed that dynamics of the search are very important. The elitist version of novelty archive with static threshold could generally not out-perform random novelty archive in clear antipattern to the fitness archives while either dynamic combinations of novelty were generally able to outperform both novelty archives.

The occasional dominance of increasing fitness over increasing novelty also seems conditioned on the structure of the environments with Cartpole rewarding similar behaviour consistently throughout it's episodes and Lunar Lander having only behavioural bottleneck at the later stage of the episode.

Originally we have also investigated Gymnasium's environment Waterworld where the fitness structure depends on other agents evolving. Ultimately it was deemed out of scope for this thesis but it is an interesting direction of these investigations left for the future works.

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List of Abbreviations

EA	Evolutionary Algorithm
ES	Evolutionary Strategy
DE	Differential Evolution
BS	Bayesian Search
IGS	Iterative grid search

A. Attachments

A.1 Full grid search charts

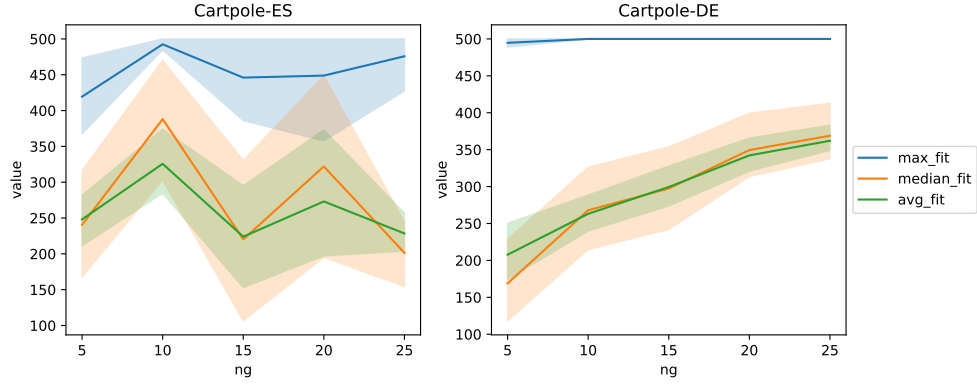


Figure A.1: Detailed fitness/generation characteristic for fitness on Cartpole env.

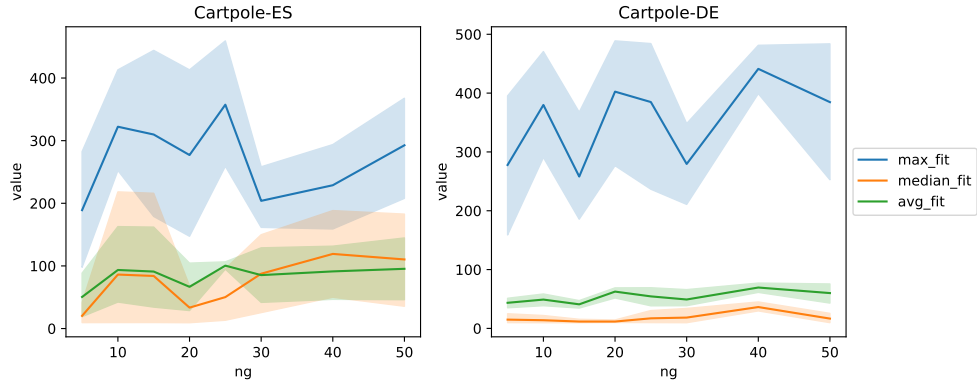


Figure A.2: Detailed fitness/generation characteristic for pure novelty on Cartpole env.

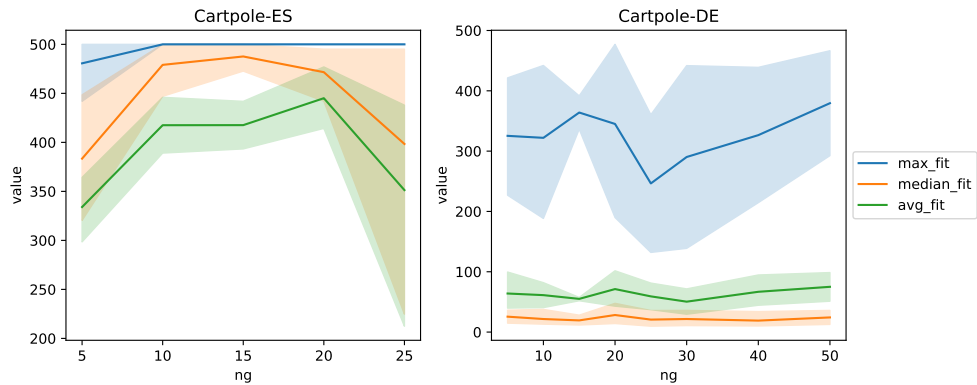


Figure A.3: Detailed fitness/generation characteristic for inc. fitness on Cartpole env.

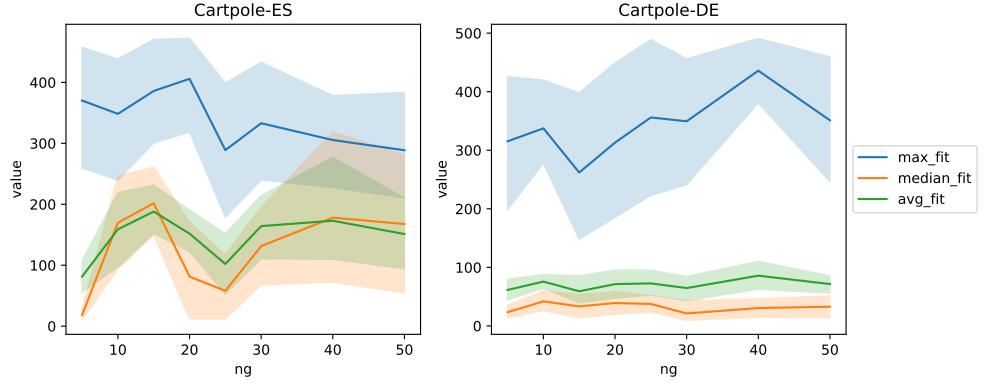


Figure A.4: Detailed fitness/generation characteristic for inc. novelty on Cartpole env.

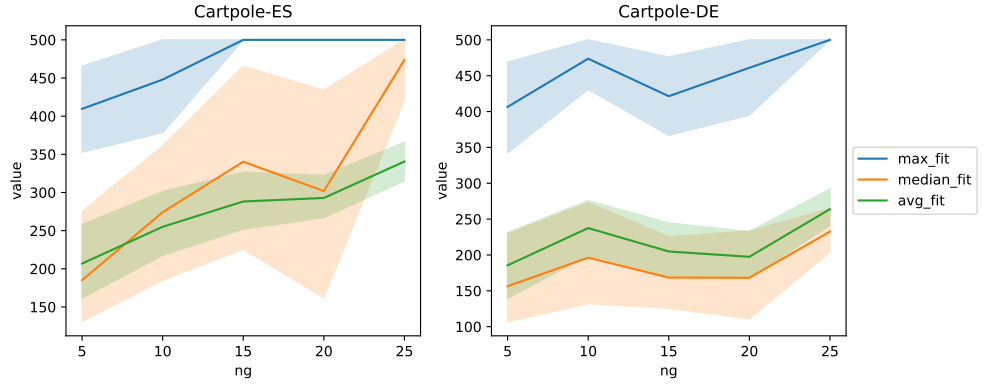


Figure A.5: Detailed fitness/generation characteristic for fit. archive on Cartpole env.

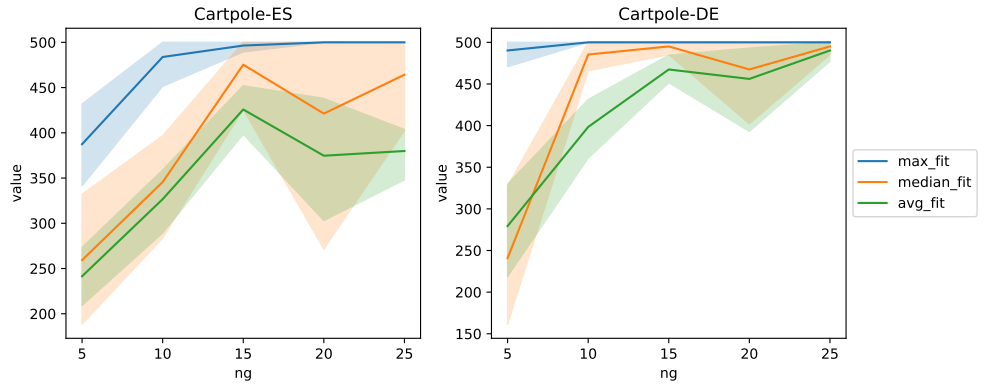


Figure A.6: Detailed fitness/generation characteristic for elite archive on Cartpole env.

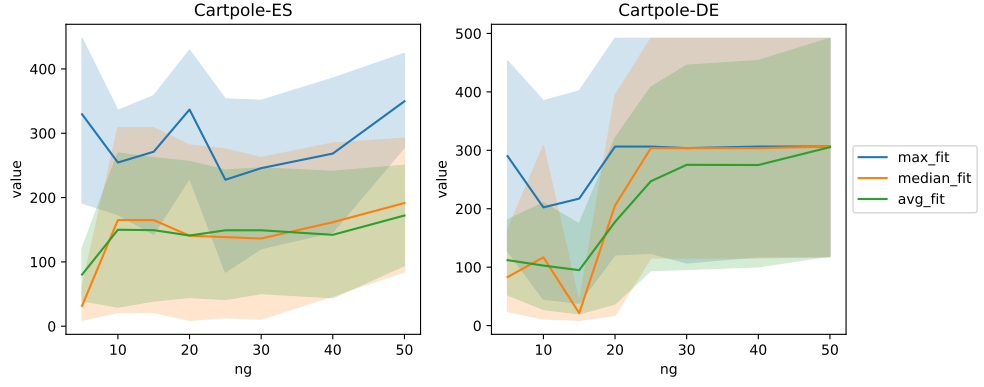


Figure A.7: Detailed fitness/generation characteristic for novelty archive on Cartpole env.

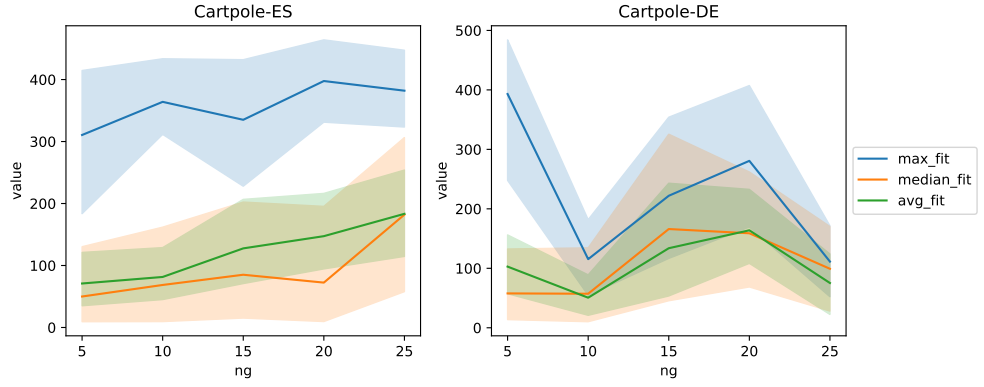


Figure A.8: Detailed fitness/generation characteristic for limit archive on Cartpole env.

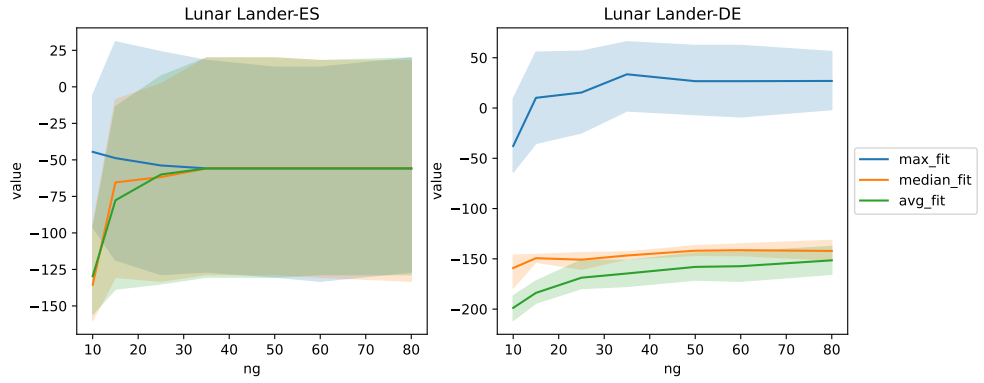


Figure A.9: Detailed fitness/generation characteristic for fitness on Lunar Lander env.

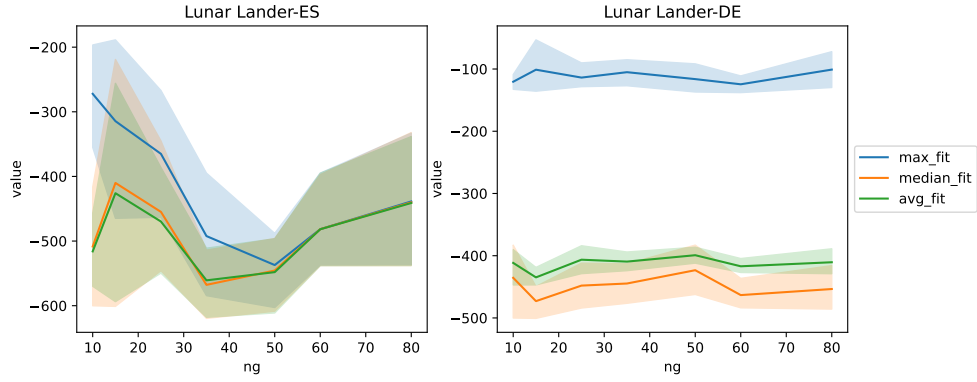


Figure A.10: Detailed fitness/generation characteristic for pure novelty on Lunar Lander env.

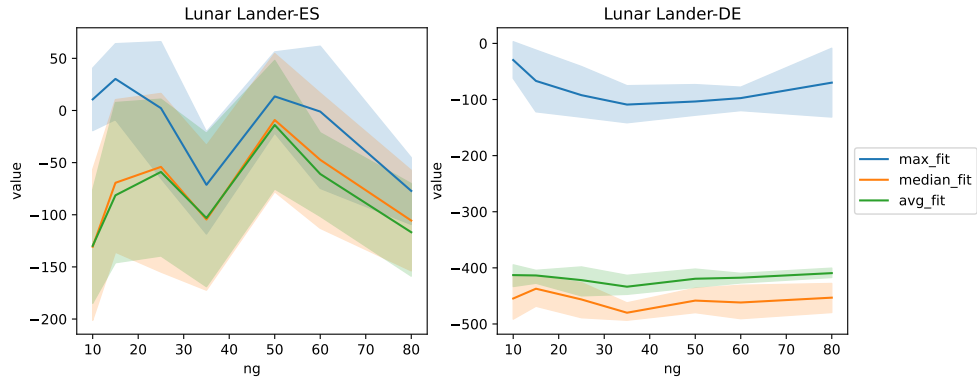


Figure A.11: Detailed fitness/generation characteristic for inc. fitness on Lunar Lander env.

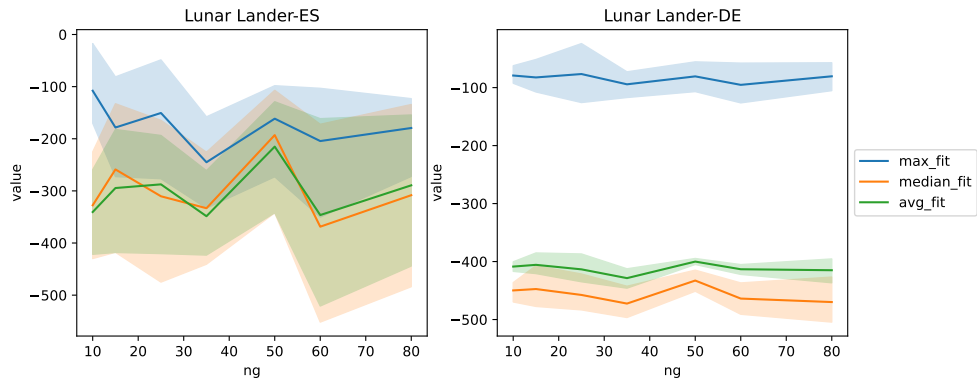


Figure A.12: Detailed fitness/generation characteristic for inc. novelty on Lunar Lander env.

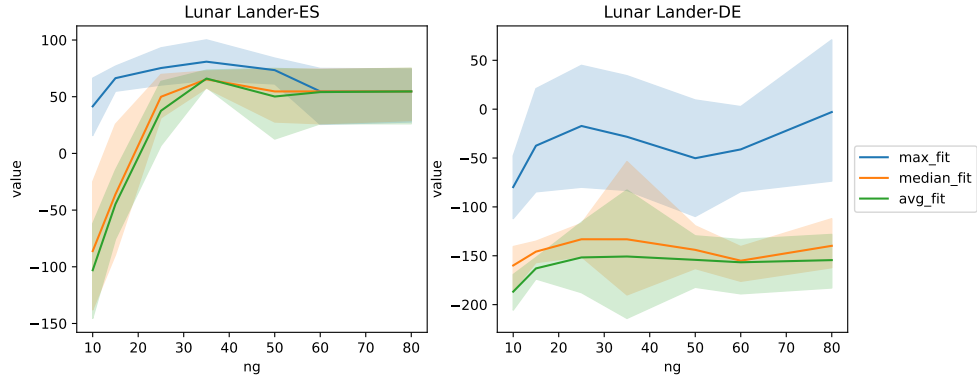


Figure A.13: Detailed fitness/generation characteristic for fit. archive on Lunar Lander env.

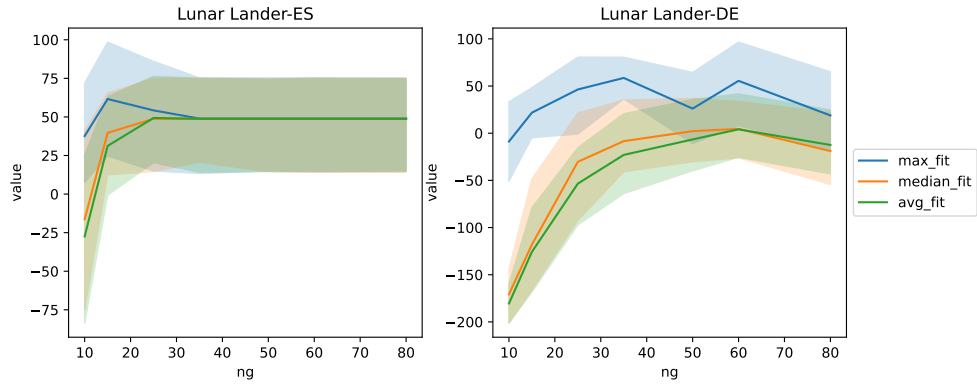


Figure A.14: Detailed fitness/generation characteristic for elite archive on Lunar Lander env.

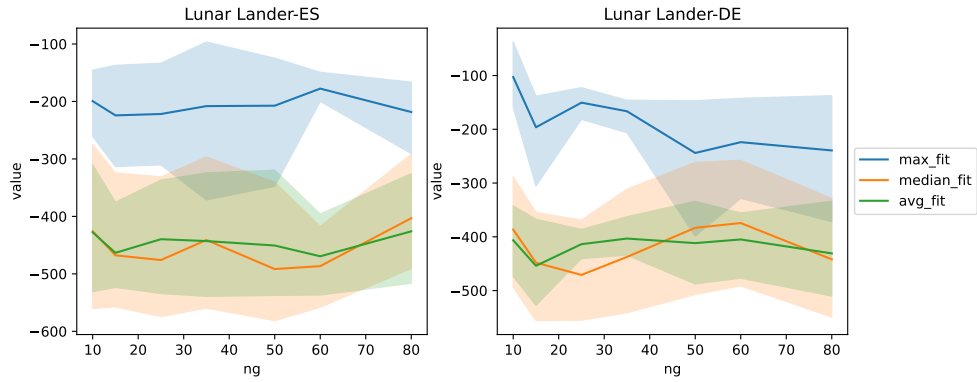


Figure A.15: Detailed fitness/generation characteristic for novetly archive on Lunar Lander env.

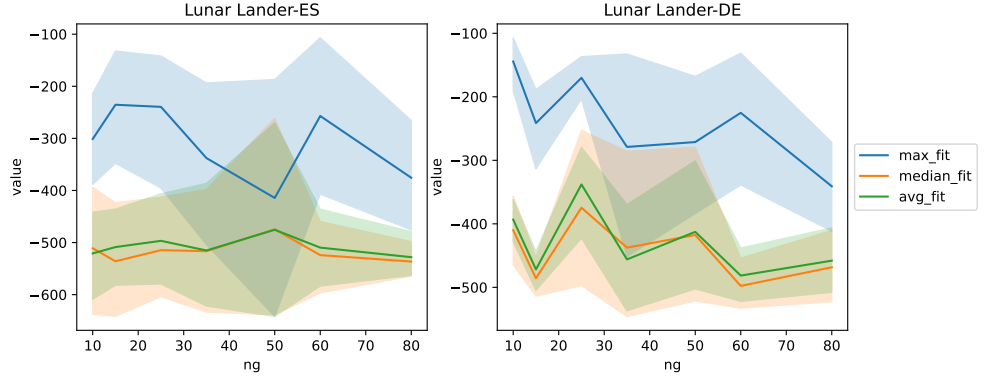


Figure A.16: Detailed fitness/generation characteristic for limit archive on Lunar Lander env.

A.2 Experiment Data Format

A.2.1 Adaptive Grid Search Protocol

```
{
  "name": "experiment_1",
  "parameters": {
    "population": 100,
    "mutation_rate": 0.1
  },
  "results": [
    0.82,
    0.91
  ]
}
```

A.2.2 Generations Grid Protocol

```
{
  "fitness": {
    "<# of generations>:{
      <seed no.>:[
        [
          [<fitness>],
          [<behavioral_vector>]
        ],
        ...
      ],
      ...
    },
    ...
  }
  "environment":<environment id>,
  "algorithm":<algorithm id>,
  "container":<container id>,
  "arguments":{
    <hyperparameter id>: <hyperparameter value>,
    ...
  }
}
```

A.2.3 Final Experiment Protocol

```
{
  "fitness": {
    <seed no.>:[
      [
        [<fitness>],
        [<behavioral_vector>]
      ],
      ...
    ],
    ...
  }
  "environment":<environment id>,
  "algorithm":<algorithm id>,
  "container":<container id>,
  "arguments":{
    <hyperparameter id>: <hyperparameter value>,
    ...
  }
}
```

A.3 Used Hardware